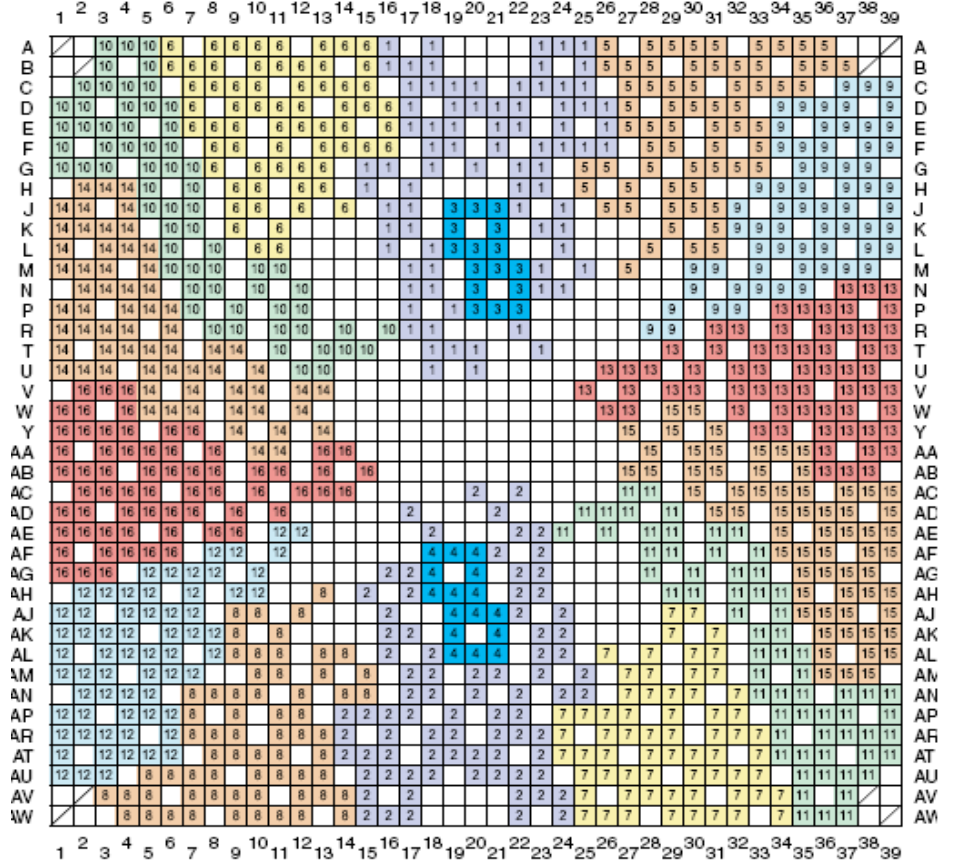
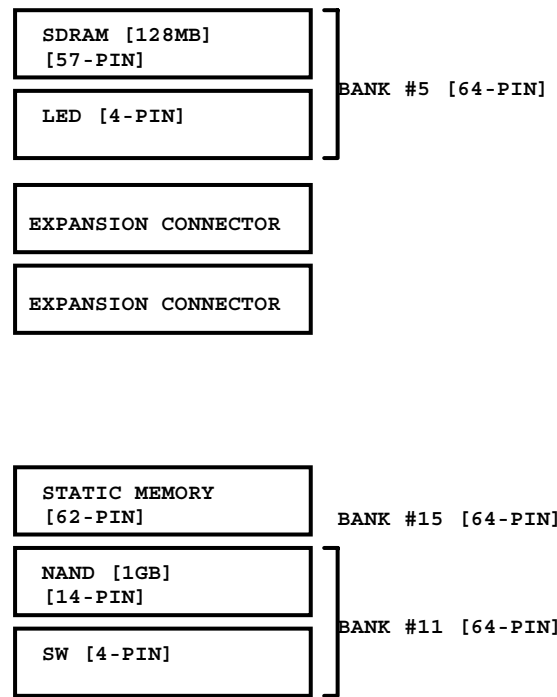
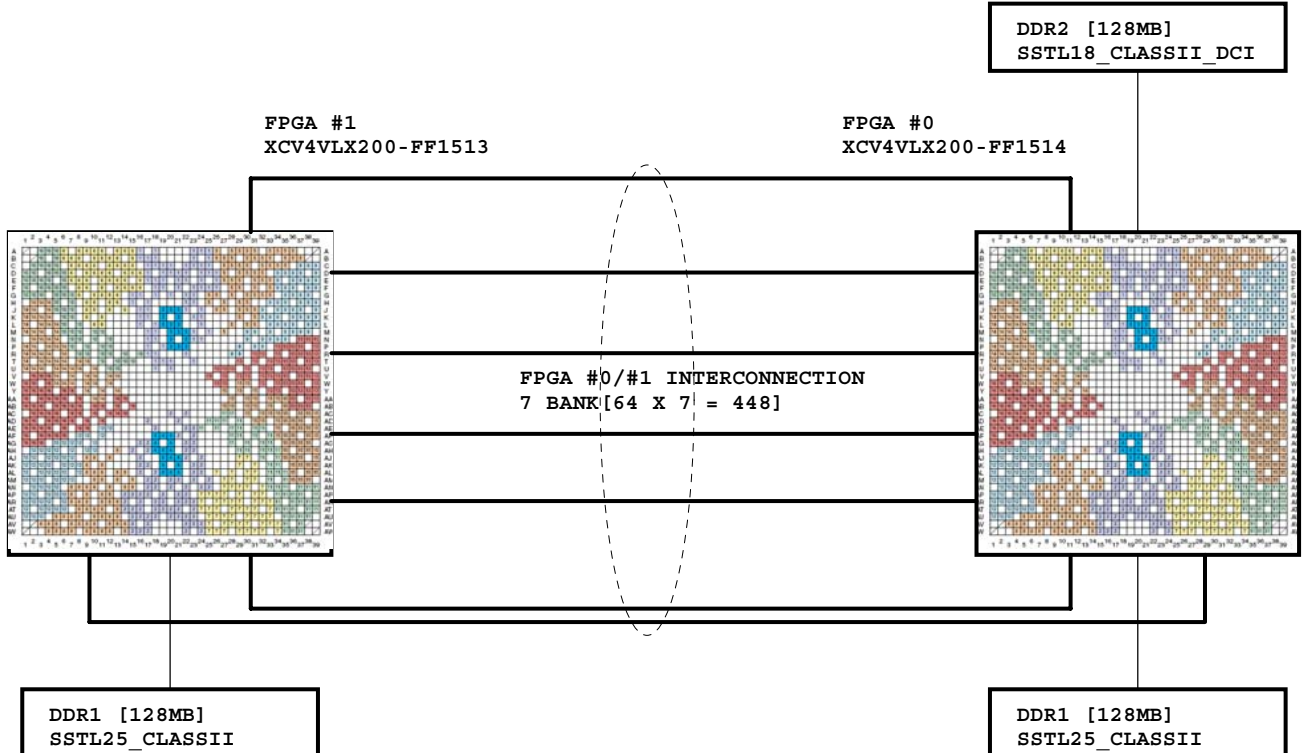
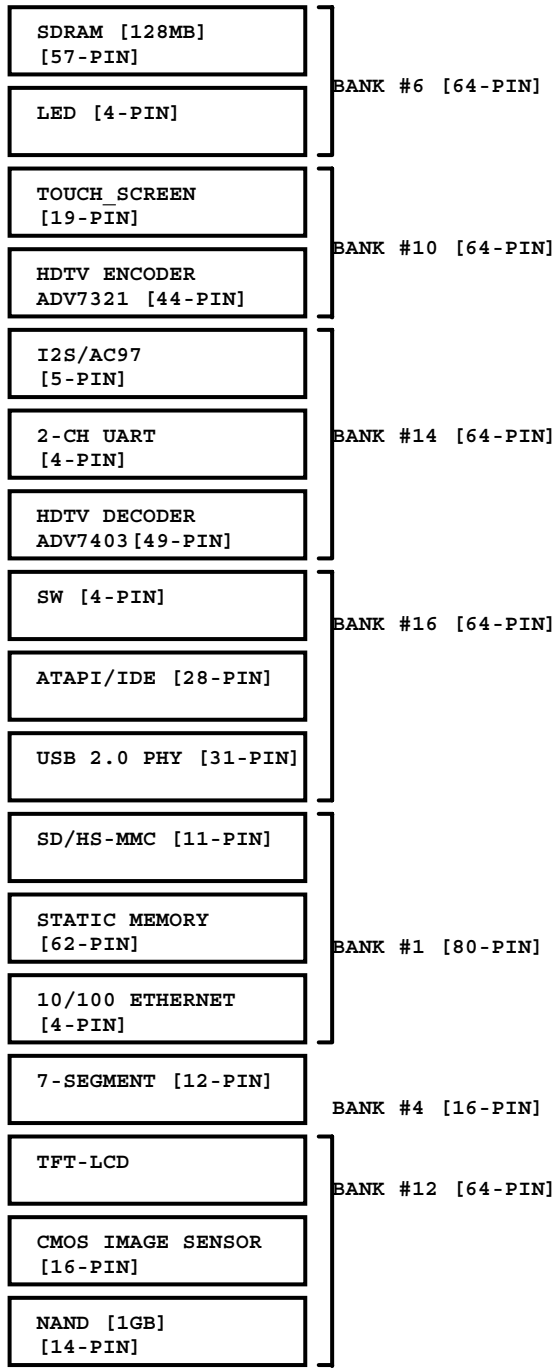


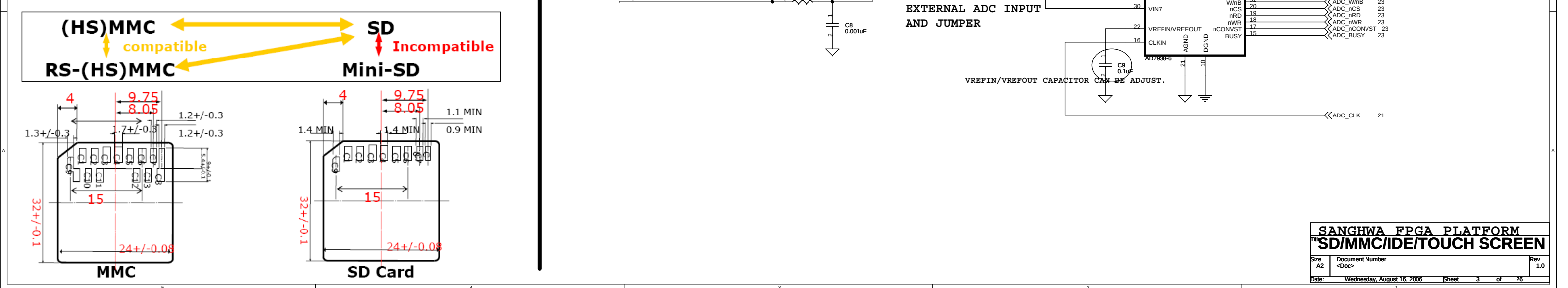
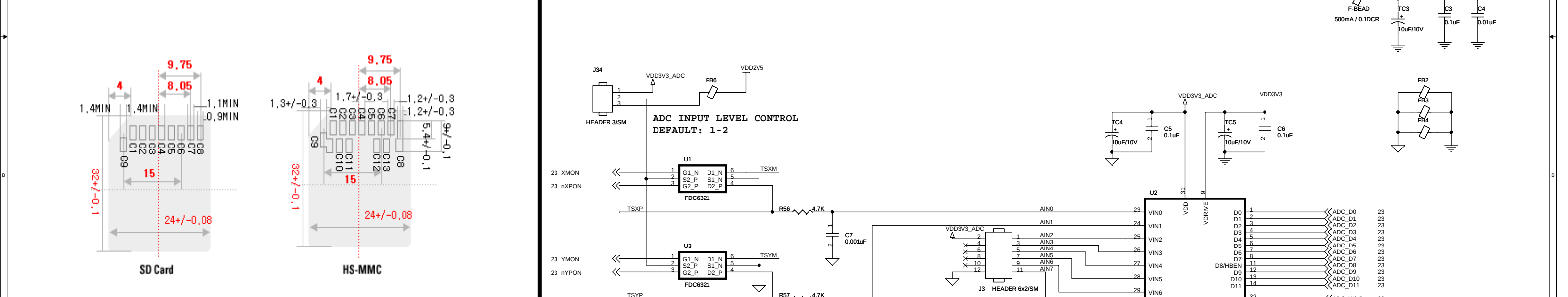
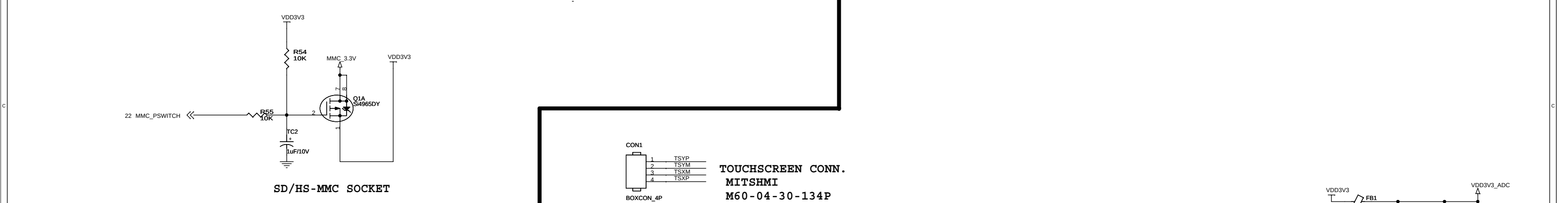
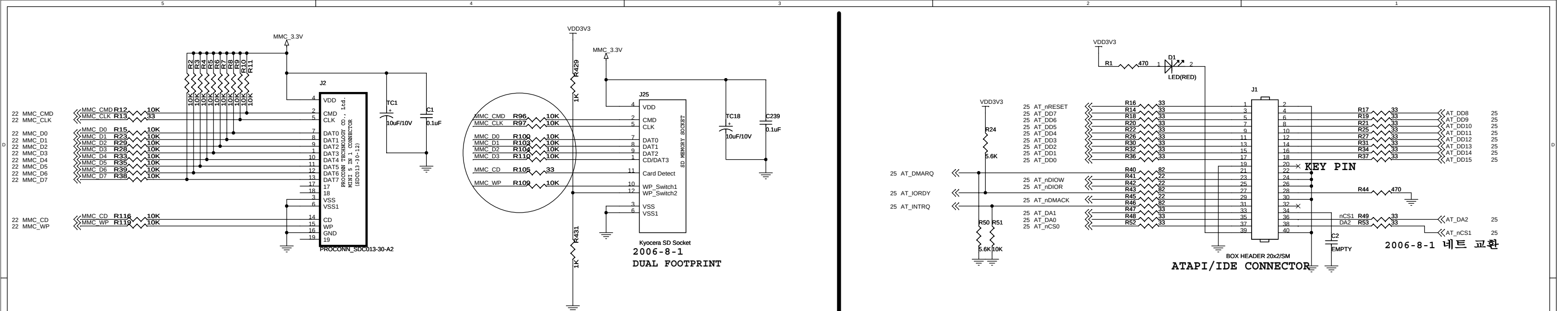
SANGHWA FPGA PLATFORM

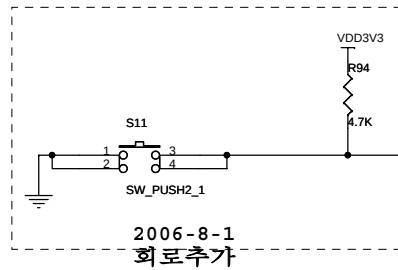
PAGE #	DESCRIPTION
P00	TITLE PAGE (THIS)
P01	SD/MMC/TOUCH
P02	SYSTEM ACE (FPGA CONFIGURATION)
P03	NONE
P04	VIDEO
P05	MEMORY BLOCK
P06	SW/AC97
P07	ETH/USB/UART
P08	CLOCK/PLL/MULTIPLEXOR
P09	SYSTEM POWER
P10	FPGA TEST MICTOR
P11	F0 CONFIG/CLOCK
P12	F0 BANK 1/2/5/6
P13	F0 BANK 7/8/9/10
P14	F0 BANK 11/12/13/14
P15	F0 BANK 15/16/GND
P16	F0 POWER VCCO
P17	F1 CONFIG/CLOCK
P18	F1 BANK 1/2/5/7
P19	F1 BANK 7/8/9/11
P20	F1 BANK 11/12/13/15
P21	F1 BANK 15/16/GND
P22	F1 POWER VCCO

2006-7-20
DESIGNED BY DDSMURF

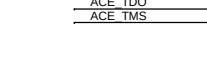
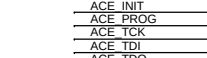
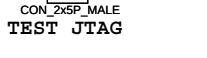
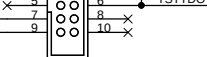
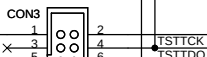
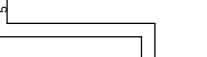
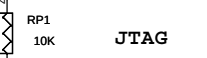
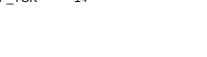
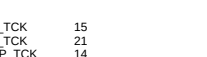
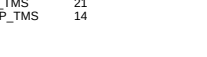
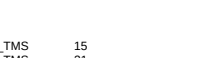
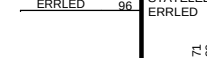
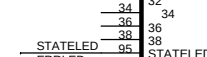
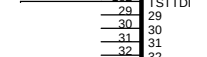
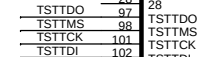
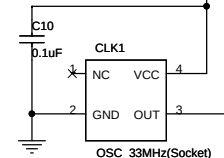
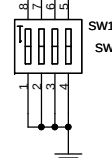
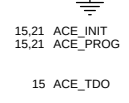
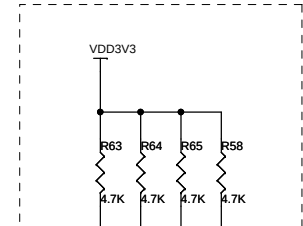
SANGHWA FPGA PLATFORM		
Title	SANGHWA FPGA PLATFORM	
Size A3	Document Number <Doc>	Rev 1.0
Date:	Monday, August 21, 2006	Sheet 1 of 26





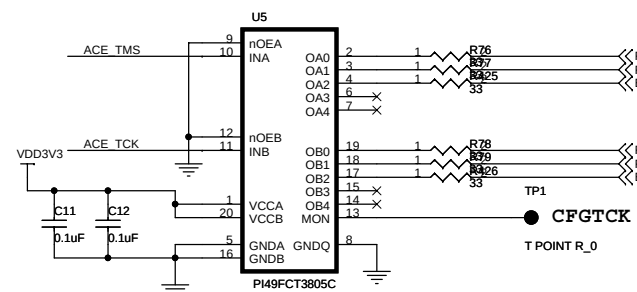


2006-8-1
회로추가



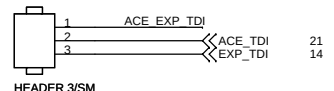
XCCACE_TQ144

ACE Module

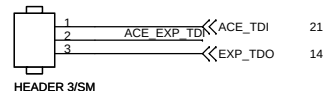


SYSTEMACE JTAG TCK/TMS BUFFER

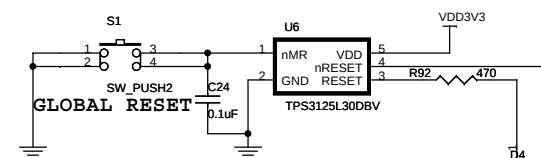
J20



J21

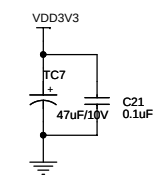
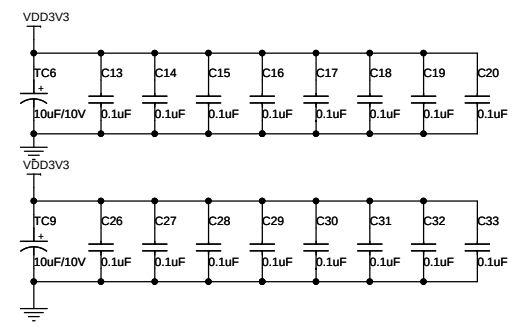
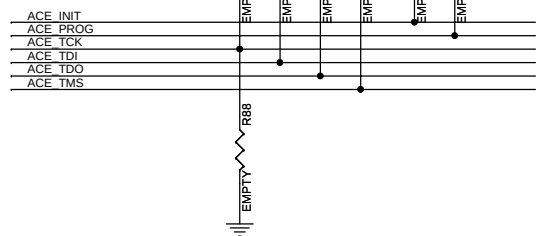
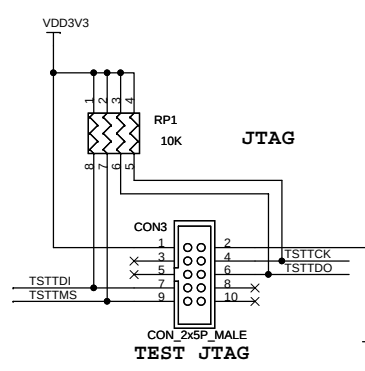
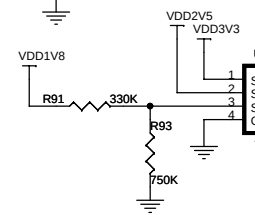
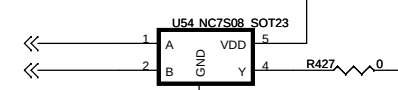


EXPANSION BOARD FPGA JTAG
J20: DEFAULT 1:2
J21: DEFAULT OPEN

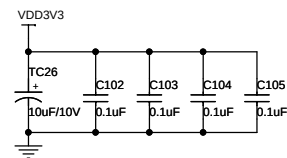
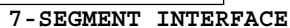
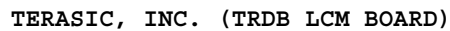
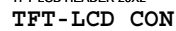
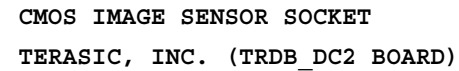
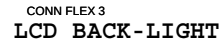
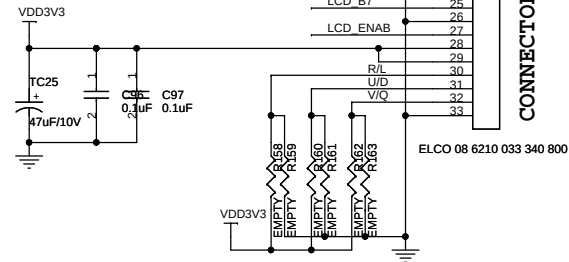


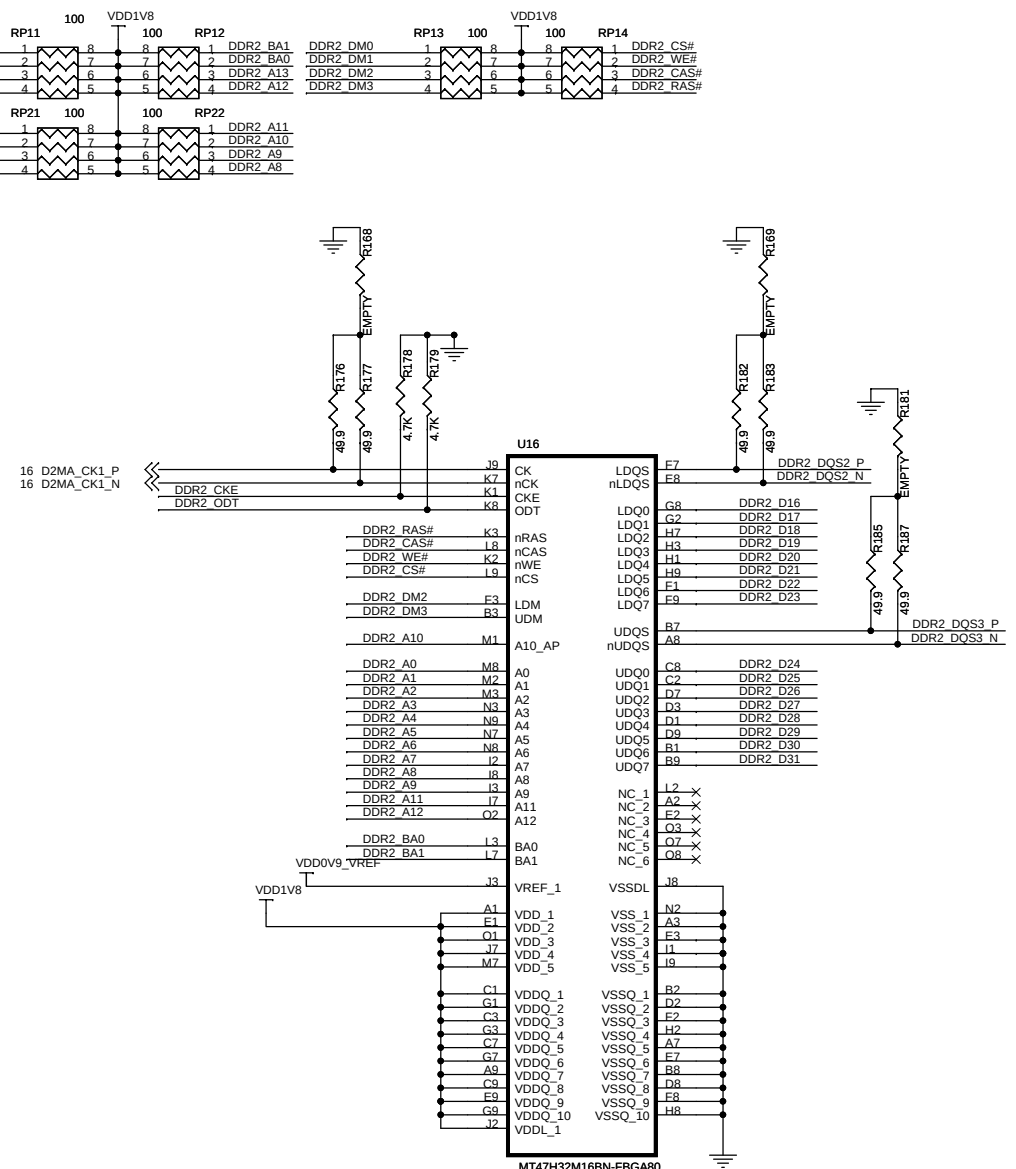
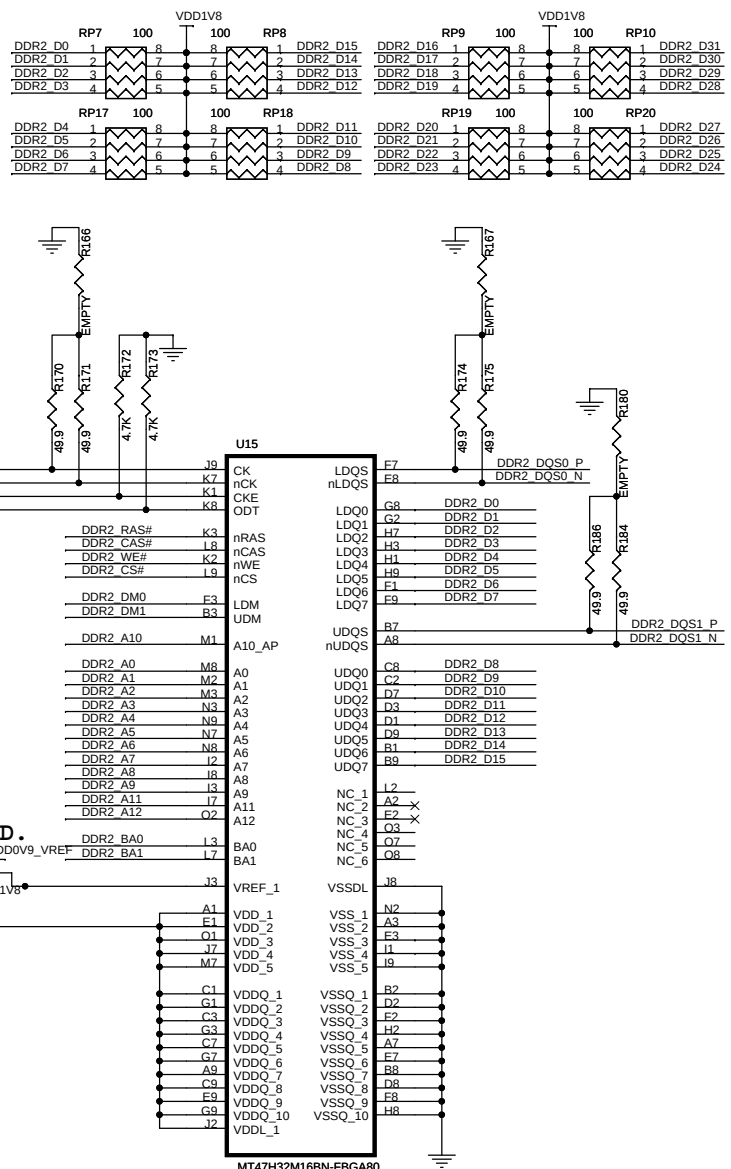
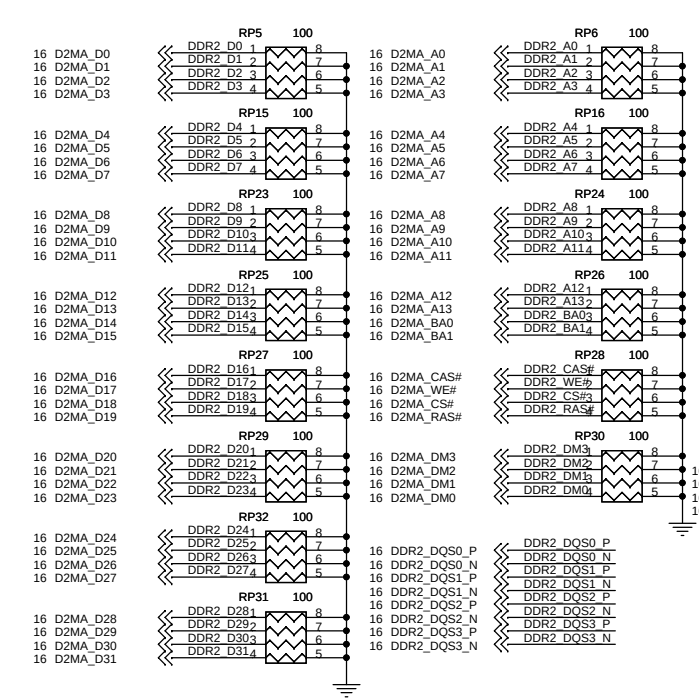
"GLOBAL RESET" SILK

15 F0_DONE
21 F1_DONE



SANGHWA FPGA PLATFORM
SYSTEM ACE





PLACE CLOSE TO DDR MEMORY DEVICES

NOTE THAT CK & DQS HAVE DIFFERENTIAL TERMINATIONS.

DQ/DQS FOR COMPONENTS DO NOT HAVE ANY TERMINATION AT THE FPGA END.

THE INTENTION IS TO RELY ON SSTL18_II DCI I/O OF FPGA FOR TERMINATION.

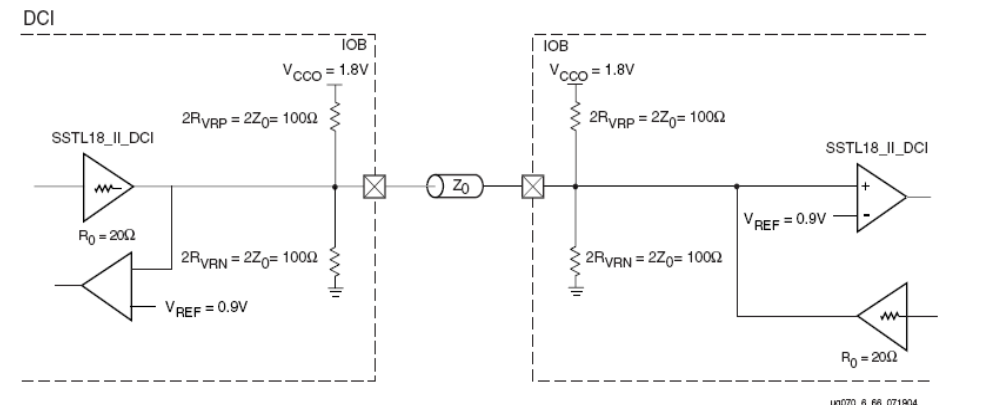
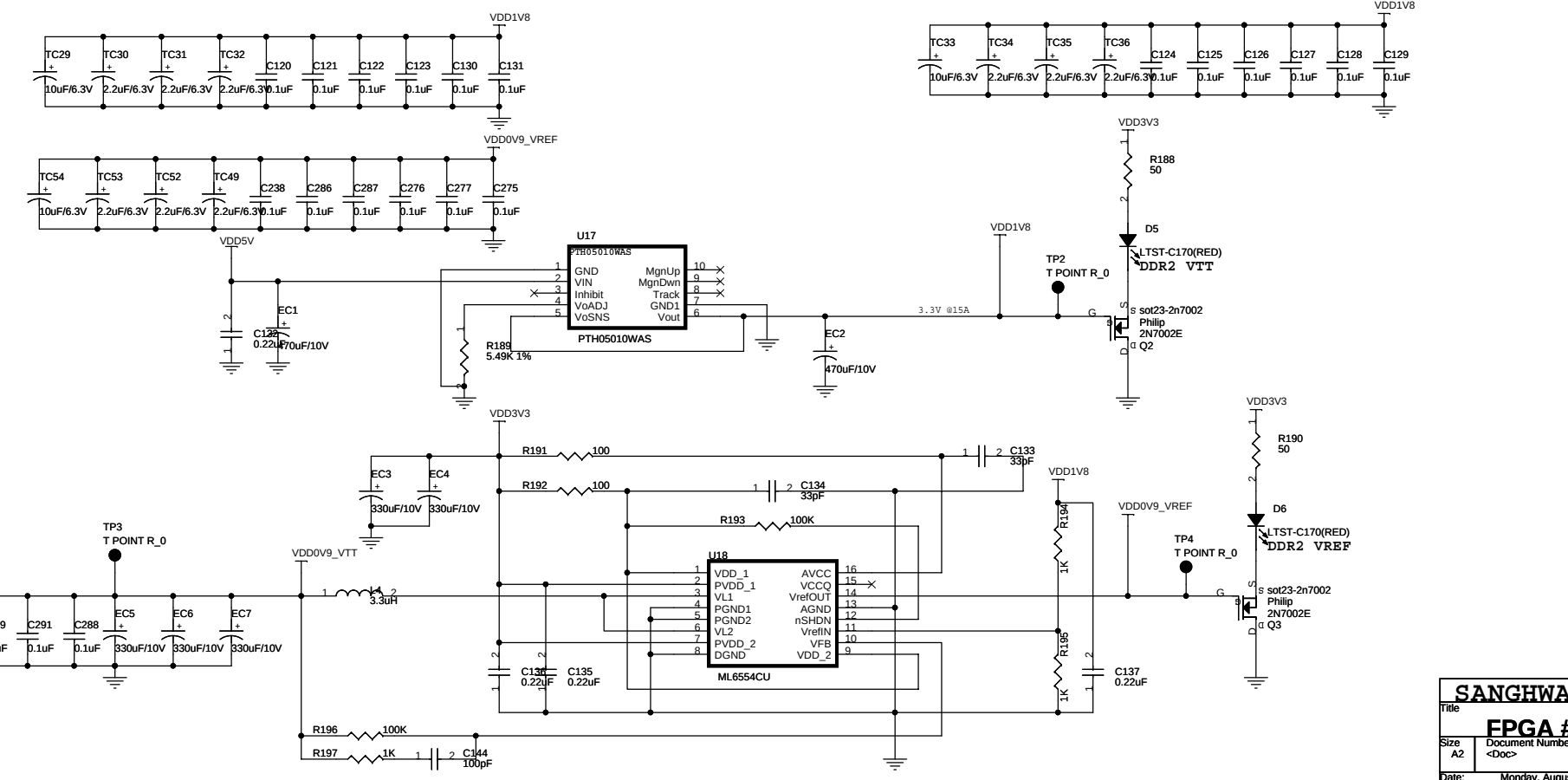
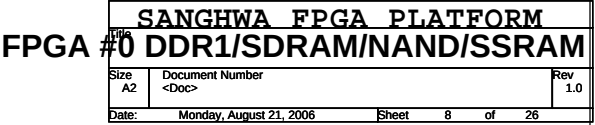
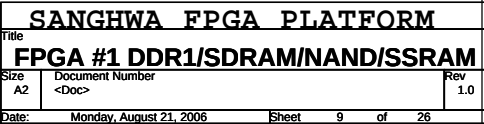


Figure 6-68: SSTL (1.8V) Class II Termination







ETH MISC SIGNALS

22 ETH_INTR
22 ETH_CS
22 ETH_RESET
22 ETH_LCLK

22 SMB_A1
9,22 SMB_A2
9,22 SMB_A3
9,22 SMB_A4
9,22 SMB_A5
9,22 SMB_A6
9,22 SMB_A7
9,22 SMB_A8
9,22 SMB_A9
9,22 SMB_A10
9,22 SMB_A11
9,22 SMB_A12
9,22 SMB_A13
9,22 SMB_A14
9,22 SMB_A15

80
81
82
83
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94

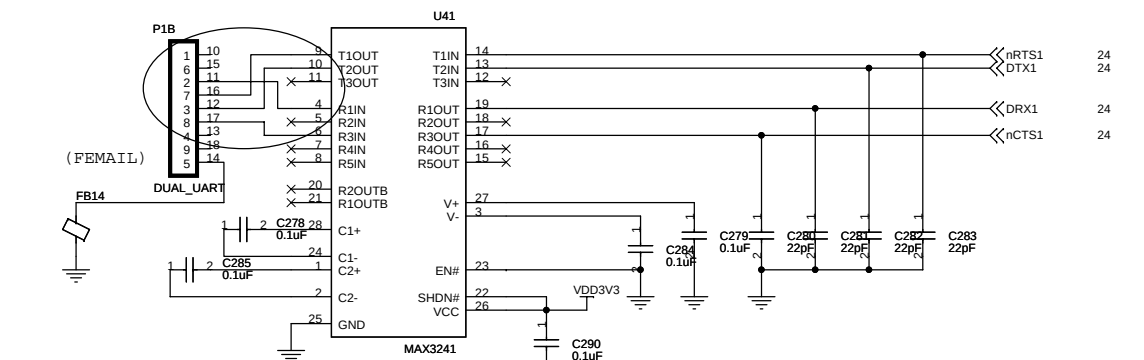
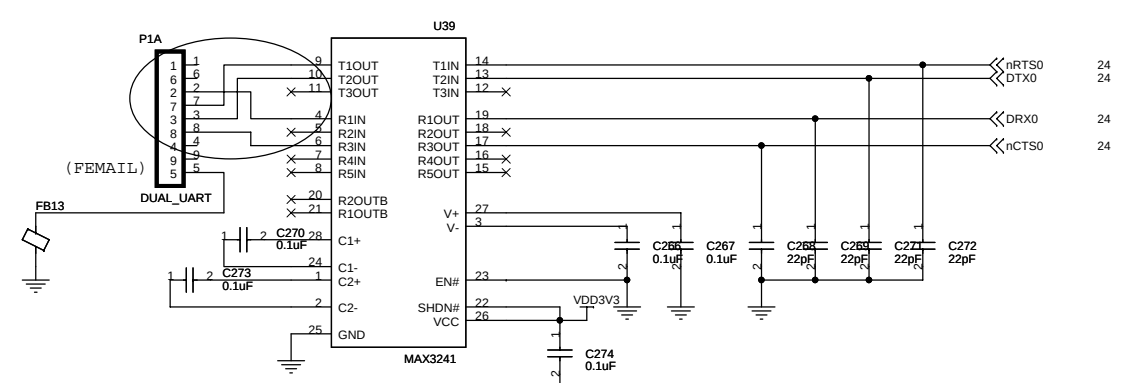
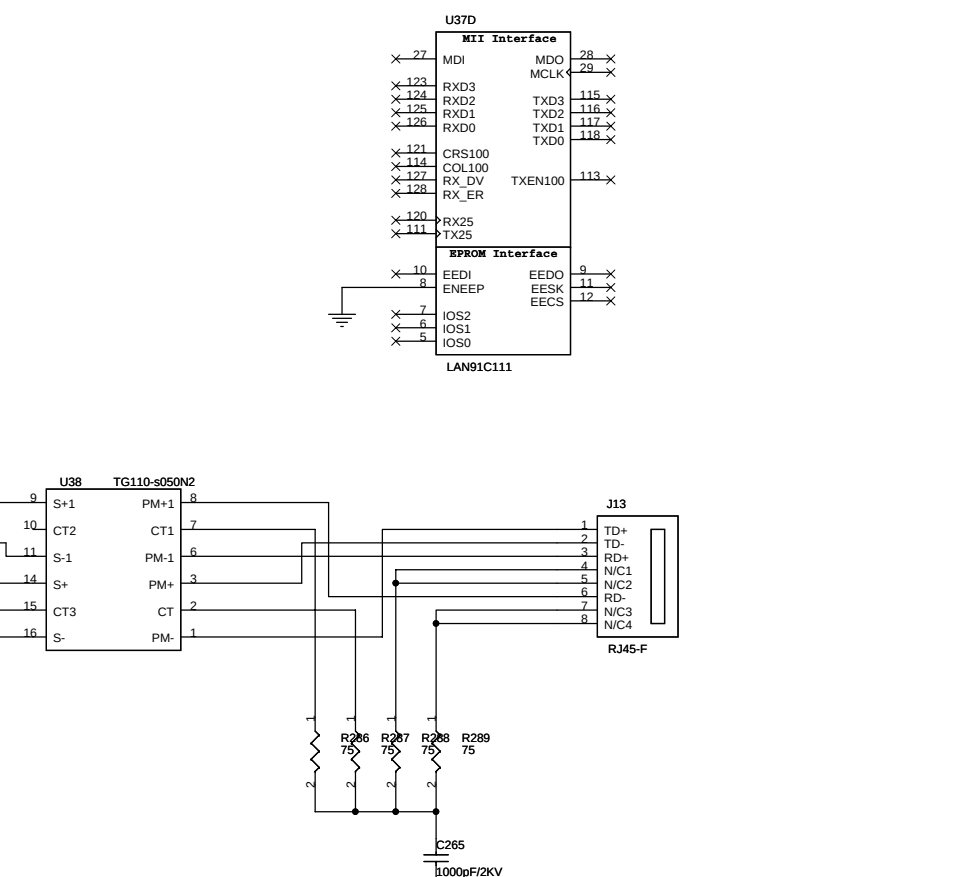
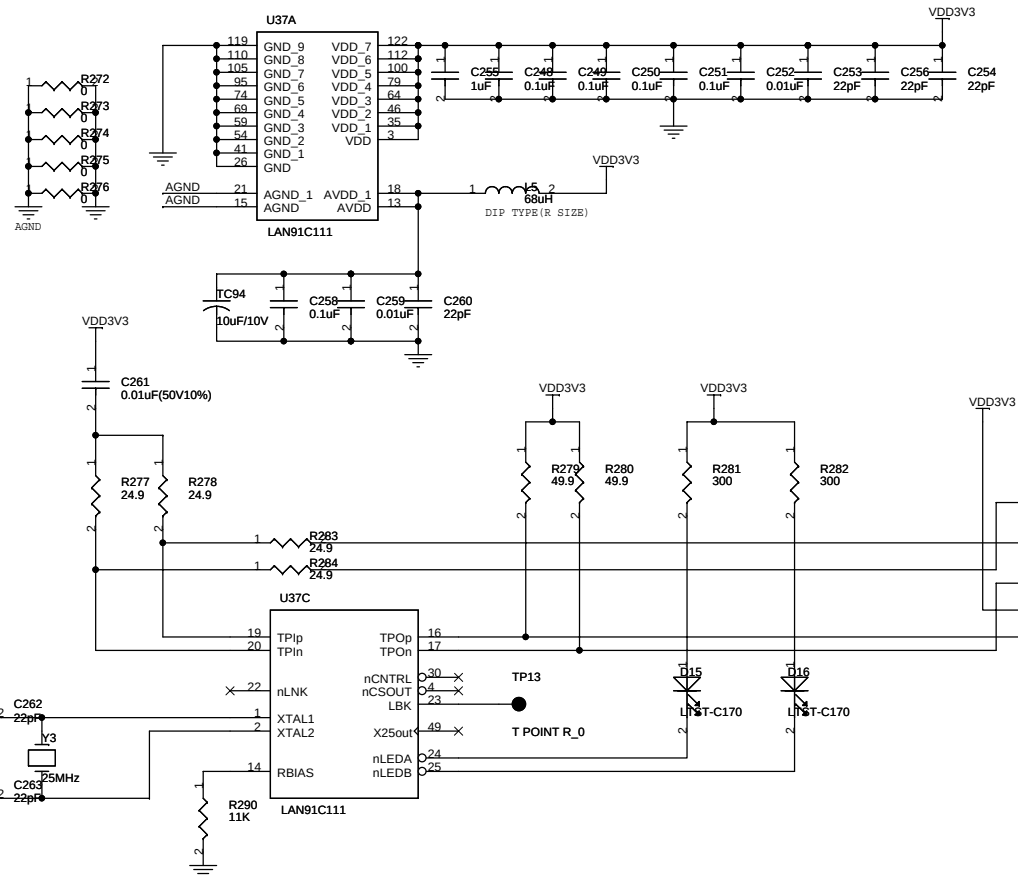
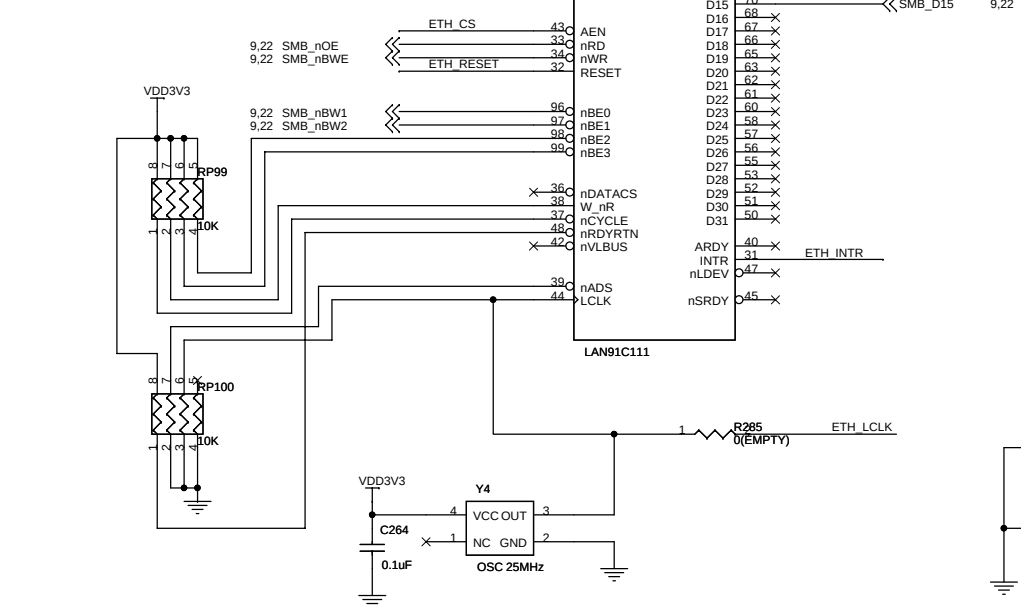
U37B

SYSTEM Interface

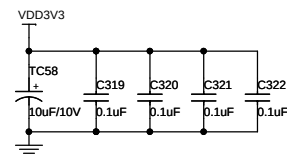
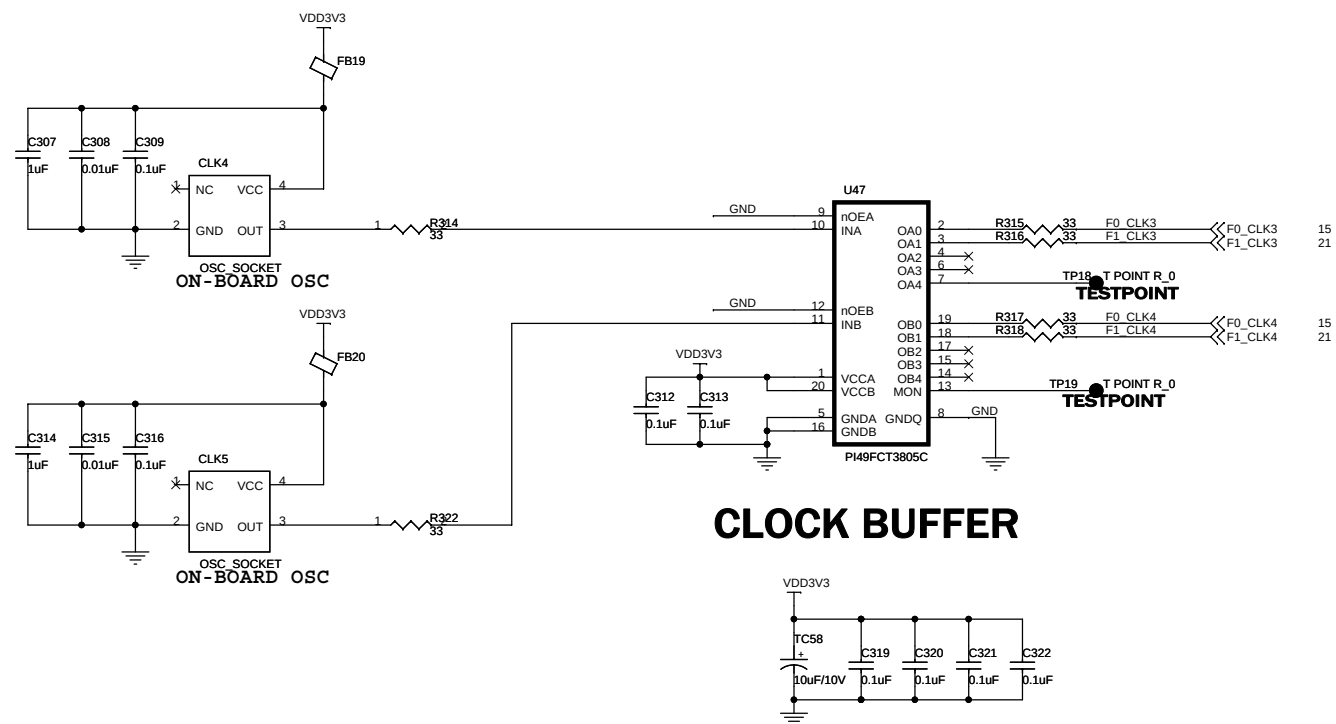
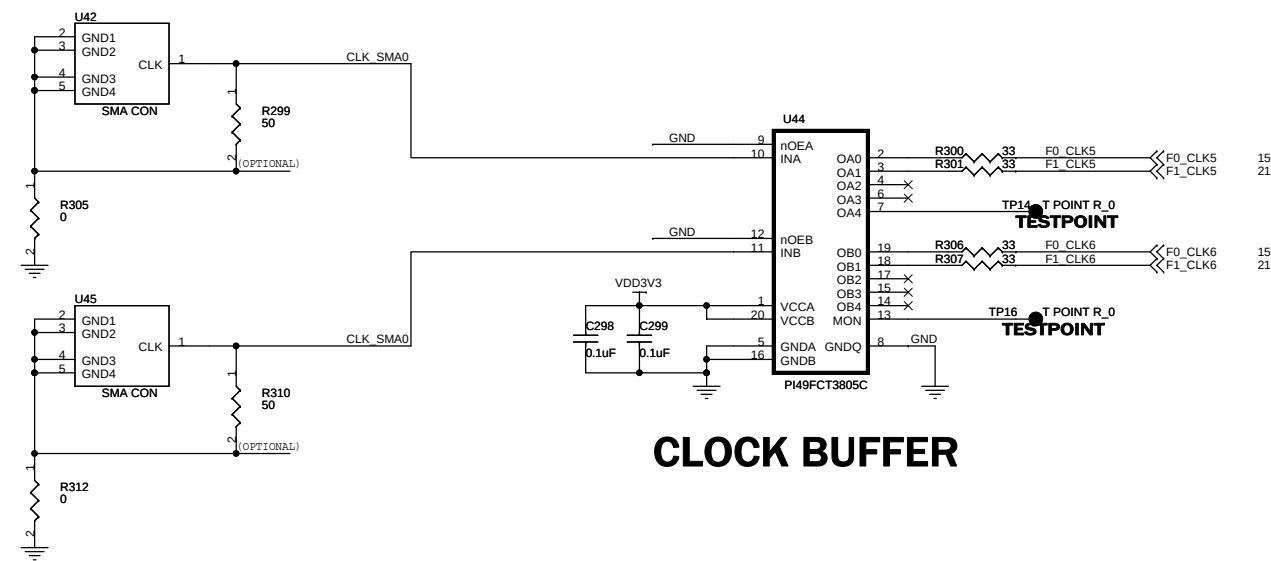
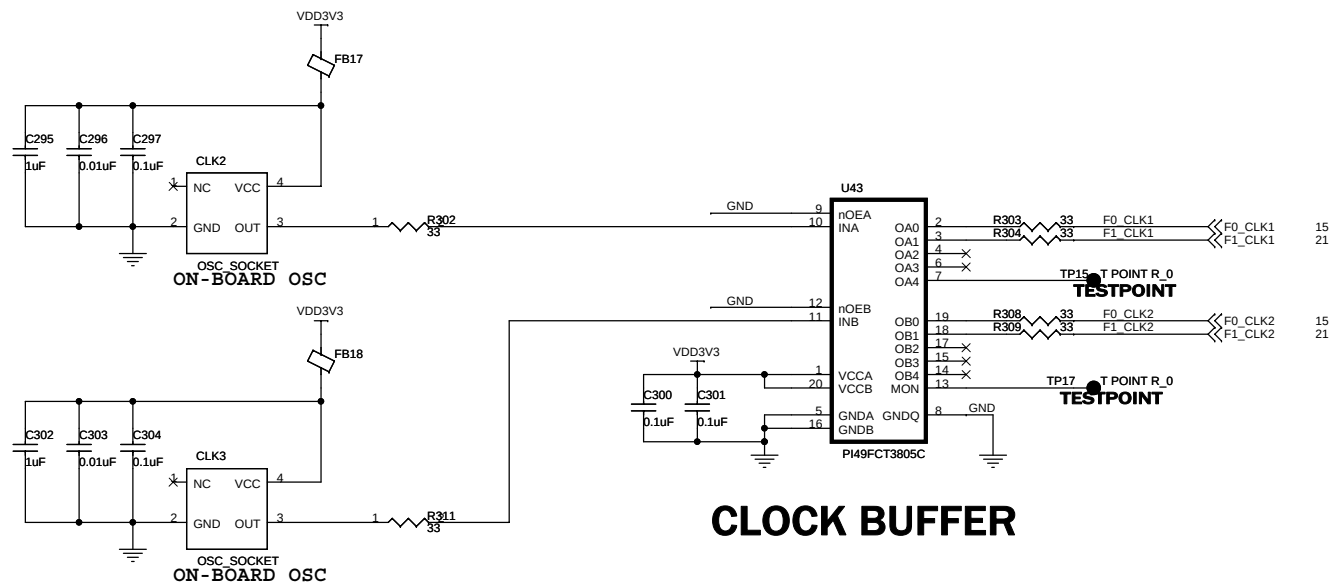
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15

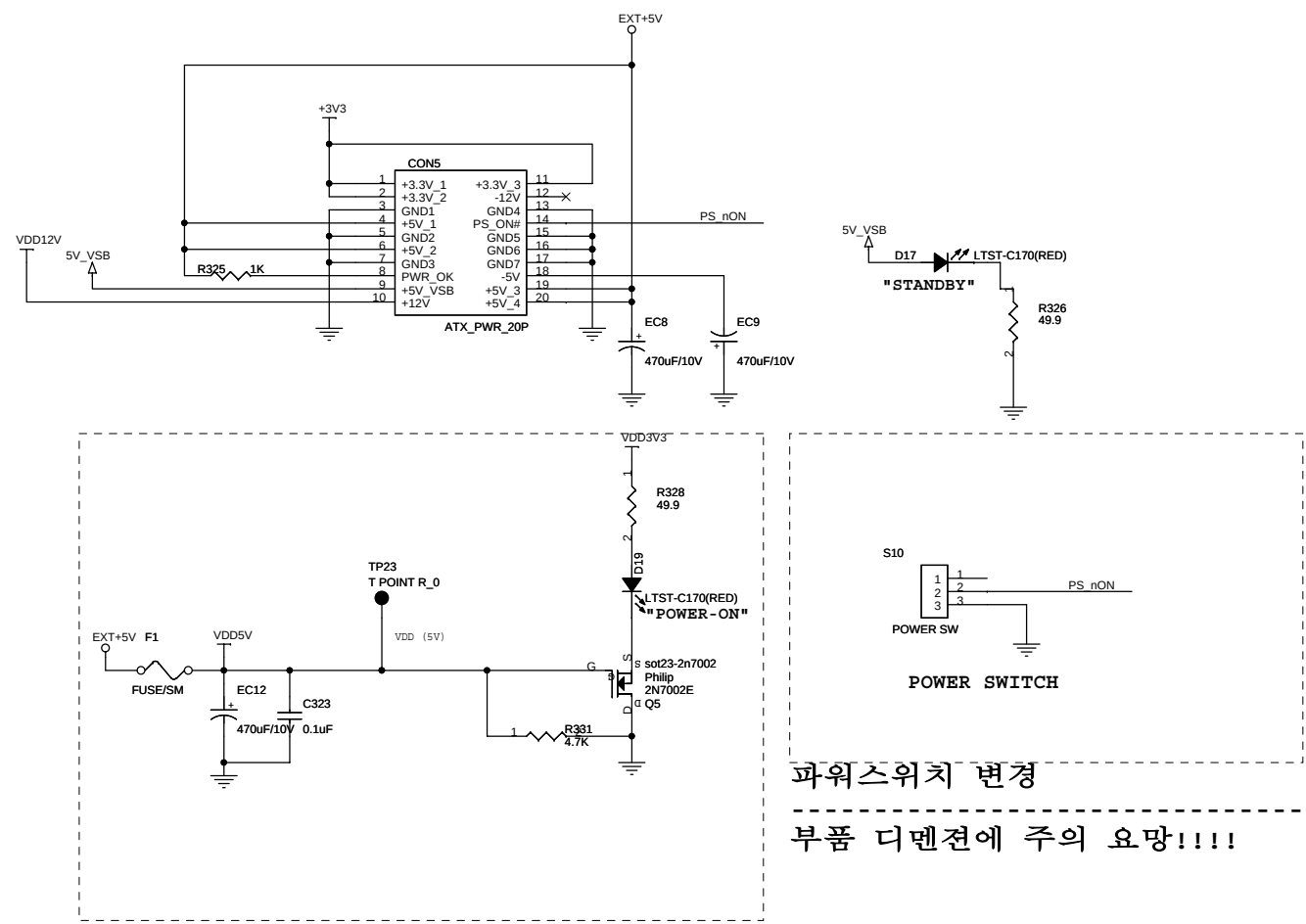
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15

SMB_D0
SMB_D1
SMB_D2
SMB_D3
SMB_D4
SMB_D5
SMB_D6
SMB_D7
SMB_D8
SMB_D9
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SMB_D11
SMB_D12
SMB_D13
SMB_D14
SMB_D15

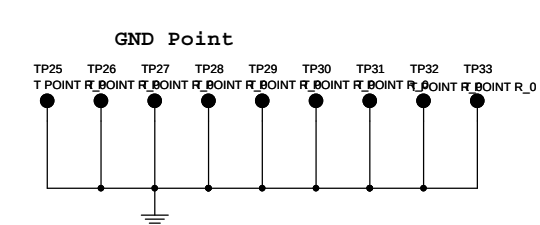
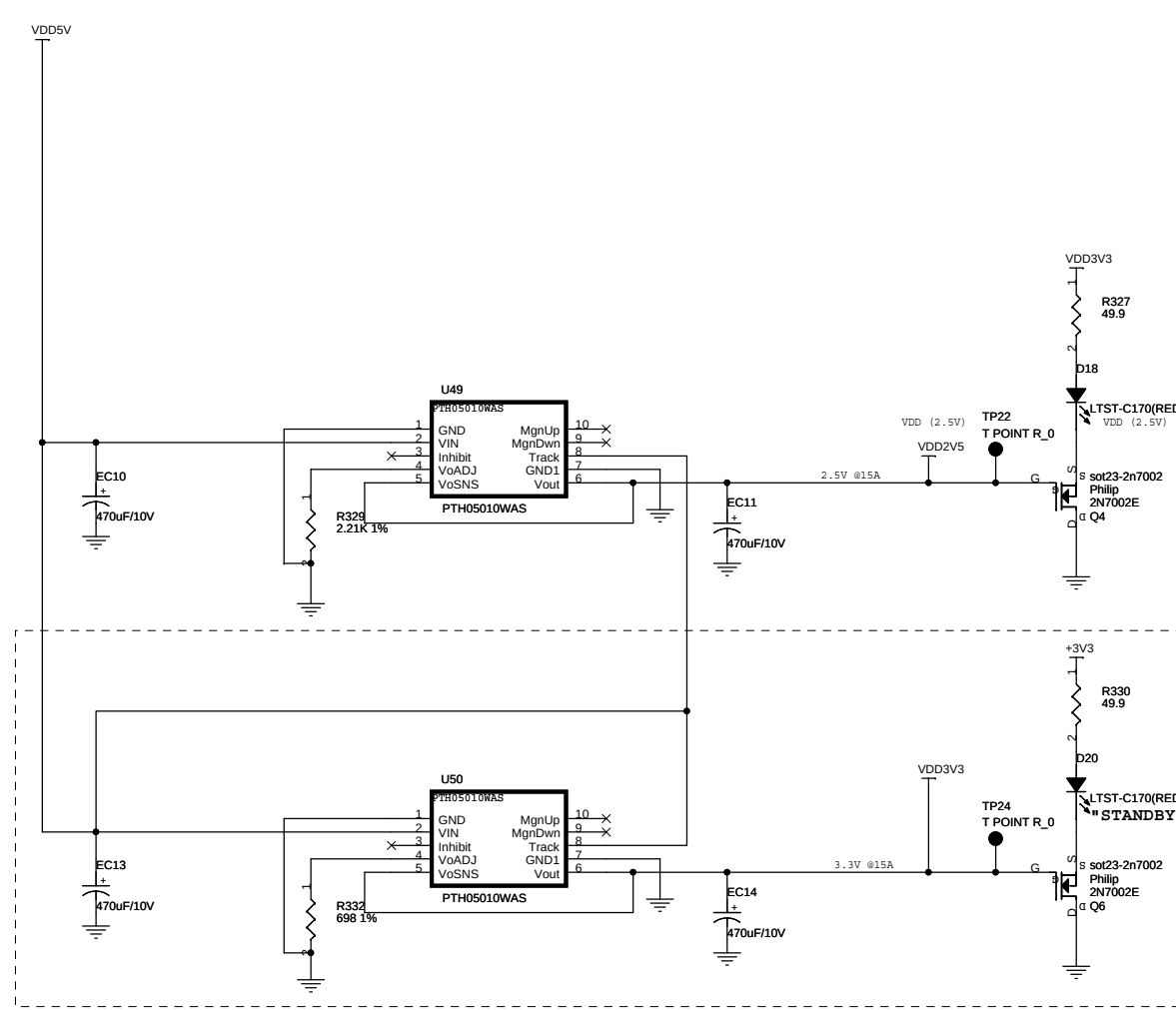


2-CH UART

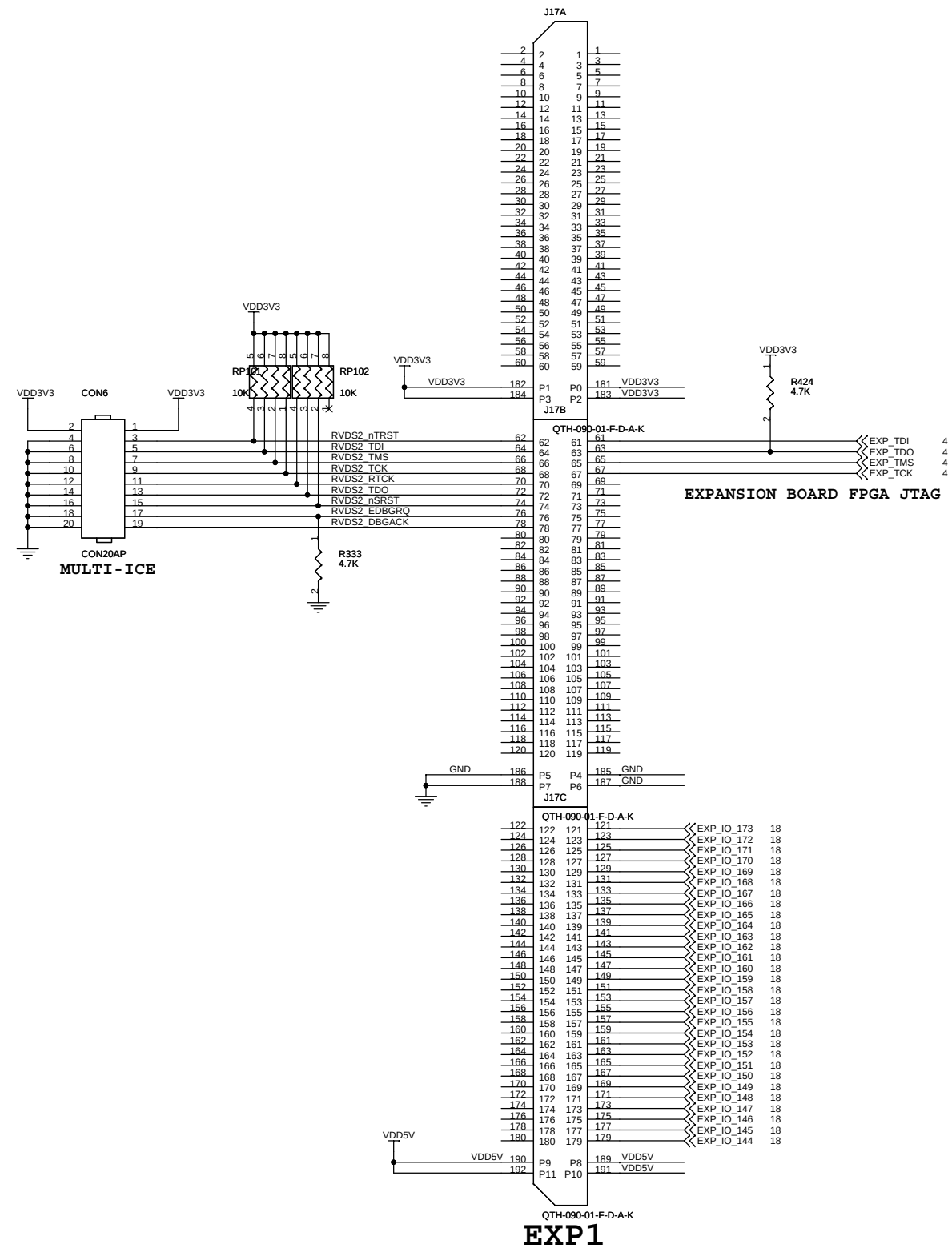
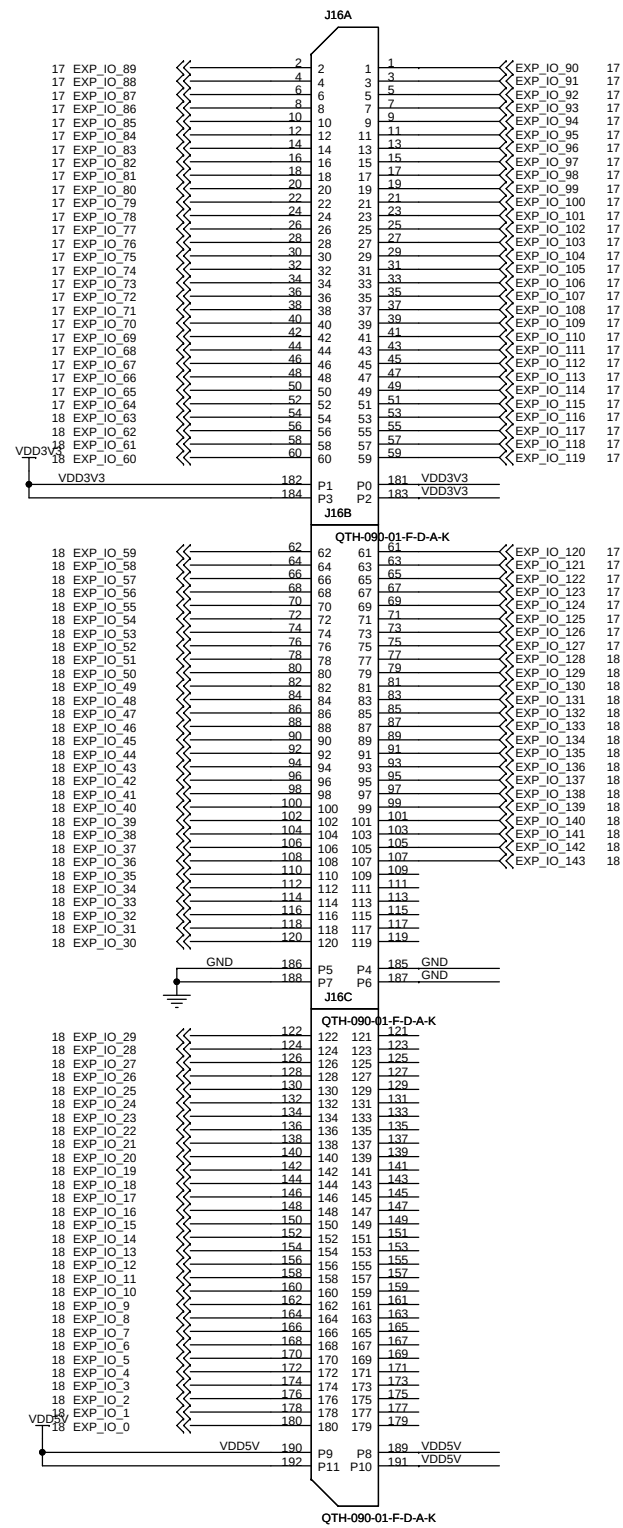


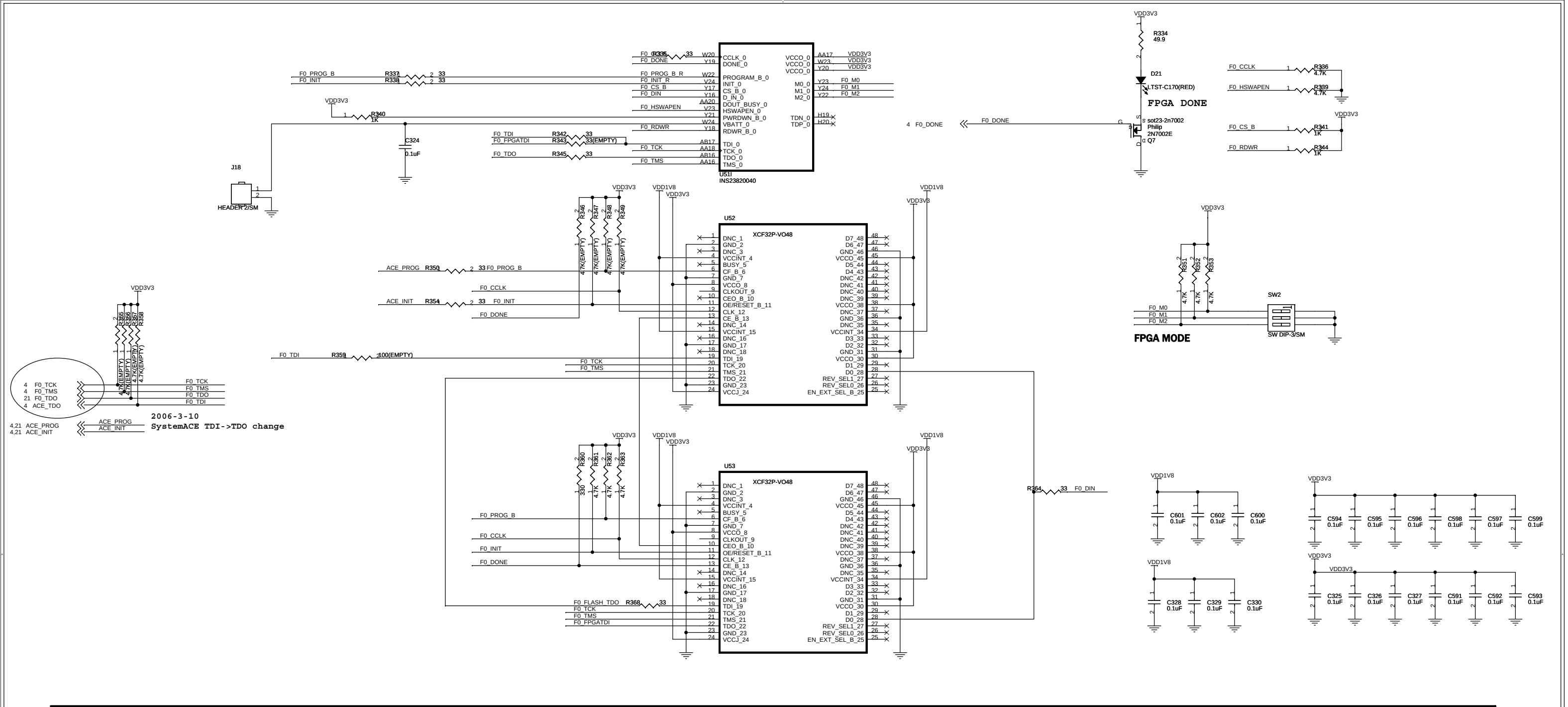


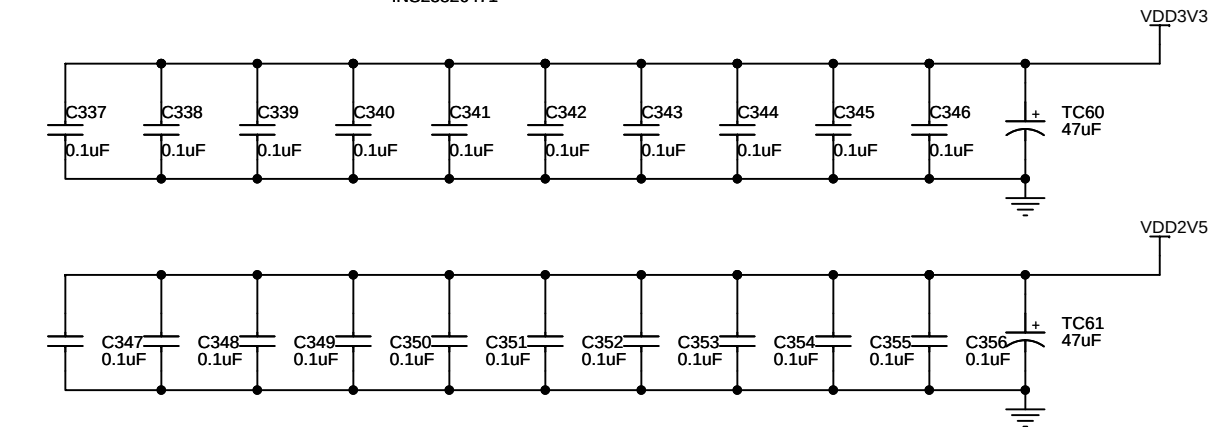
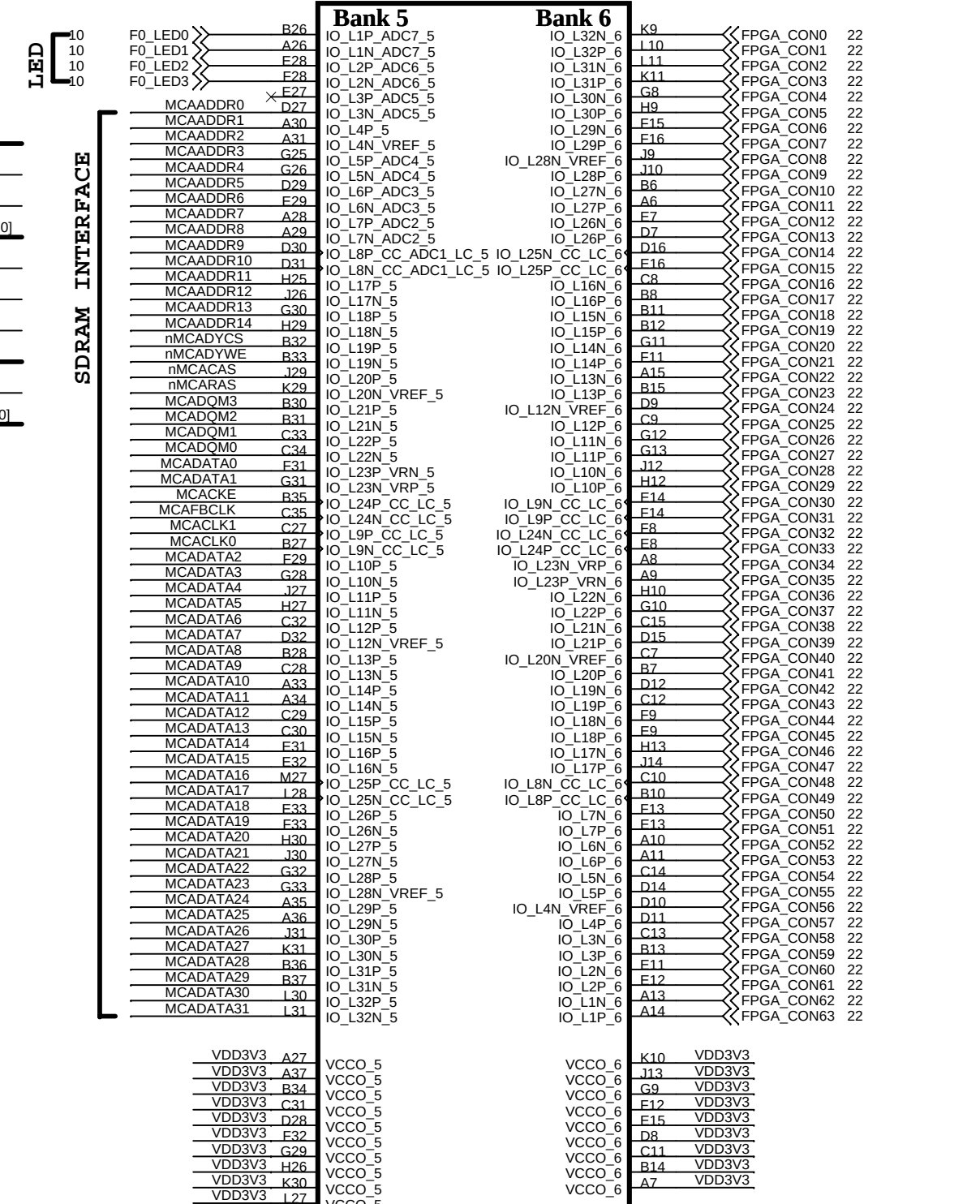
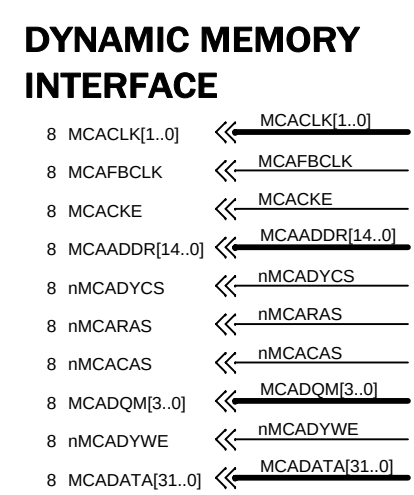
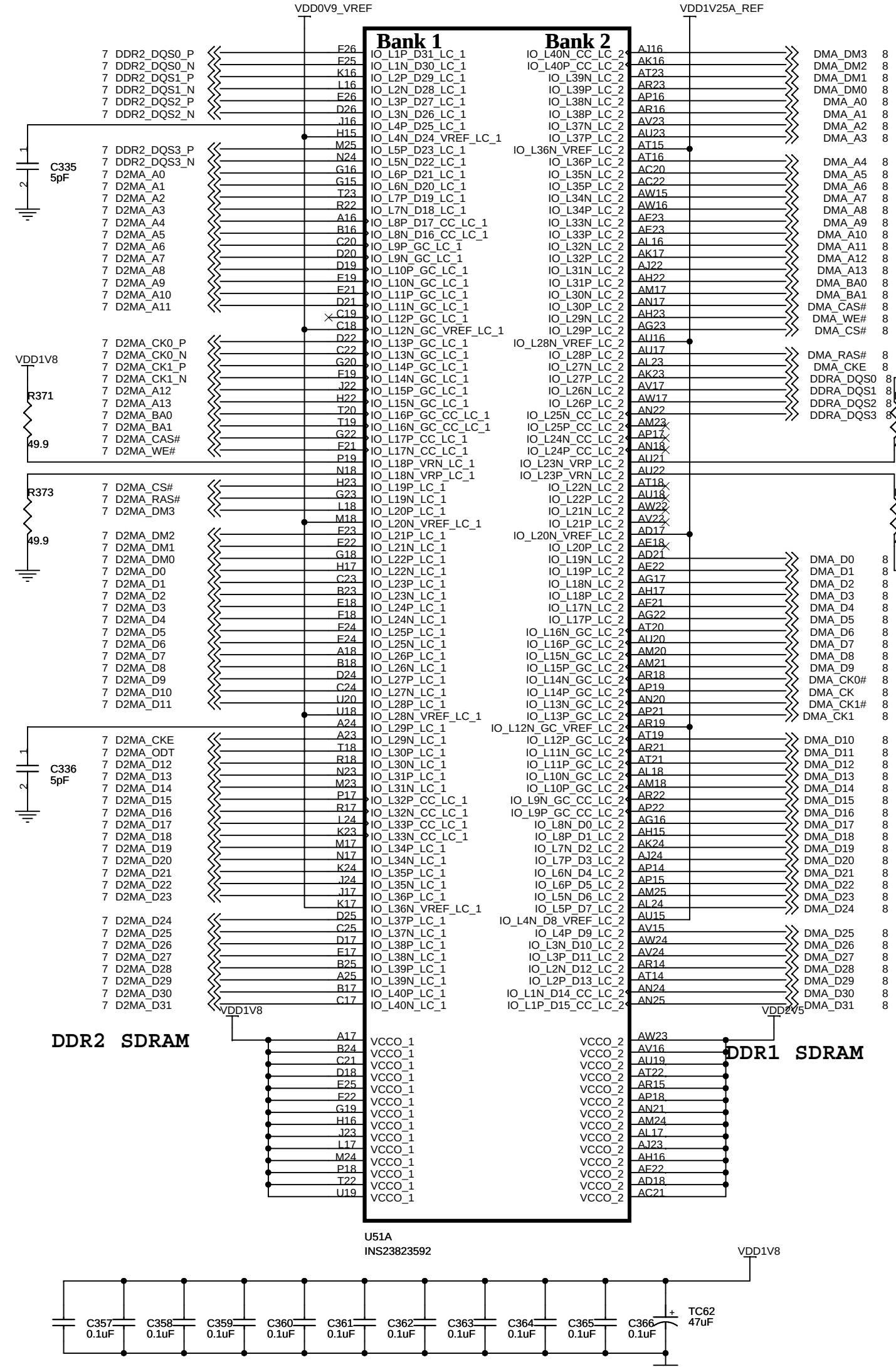
파워스위치 변경
부품 디멘전에 주의 요망!!!!



EXPANSION SLOT FOR FPGA #0

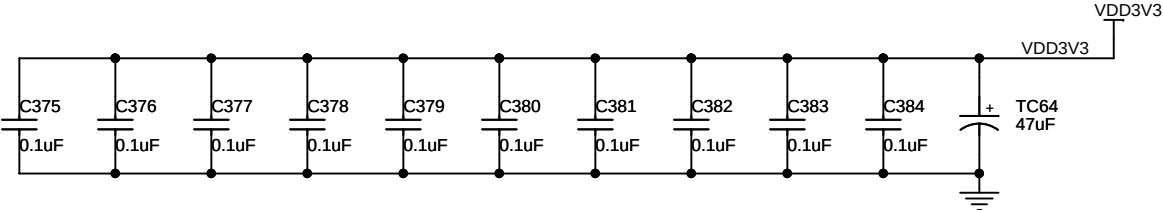
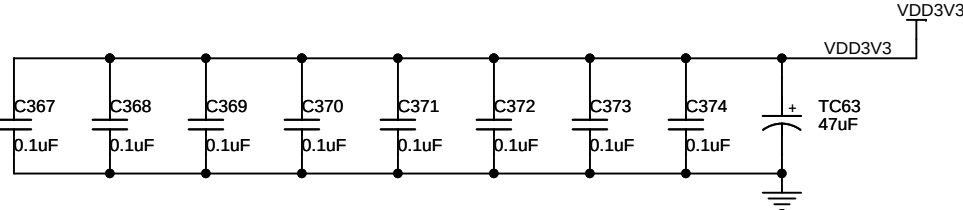






FPGA INTERCONNECTION

Bank 7			Bank 8			Bank 9			Bank 10				
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23	FPGA_CON385	AR33	IO_L1N_7	IO_L32P_8	AW11	FPGA_CON321	23	D35	IO_L1N_9	IO_L32P_10	U12	FPGA_CON65	23
23	FPGA_CON386	AI30	IO_L2P_7	IO_L31N_8	AV14	FPGA_CON322	23	C37	IO_L2P_9	IO_L31N_10	R8	FPGA_CON66	23
23	FPGA_CON387	AK31	IO_L2N_7	IO_L31P_8	AW14	FPGA_CON323	23	C38	IO_L2N_9	IO_L31P_10	R9	FPGA_CON67	23
23	FPGA_CON388	AM30	IO_L3P_7	IO_L30N_8	AU10	FPGA_CON324	23	F34	IO_L3P_9	IO_L30N_10	P7	FPGA_CON68	23
23	FPGA_CON389	AL30	IO_L3N_7	IO_L30P_8	AV10	FPGA_CON325	23	F34	IO_L3N_9	IO_L30P_10	N7	FPGA_CON69	23
23	FPGA_CON390	AM31	IO_L4P_7	IO_L29N_8	AR13	FPGA_CON326	23	F35	IO_L4P_9	IO_L29N_10	M6	FPGA_CON70	23
23	FPGA_CON391	AI31	IO_L4N_VREF_7	IO_L29P_8	AR11	FPGA_CON327	23	G35	IO_L4N_VREF_9	IO_L29P_10	N8	FPGA_CON71	23
23	FPGA_CON392	AP24	IO_L5P_7	IO_L28N_VREF_8	AT11	FPGA_CON328	23	D36	IO_L5P_9	IO_L28N_VREF_10	P9	FPGA_CON72	23
23	FPGA_CON393	AR24	IO_L5N_7	IO_L28P_8	AM13	FPGA_CON329	23	D37	IO_L5N_9	IO_L28P_10	M7	FPGA_CON73	23
23	FPGA_CON394	AP32	IO_L6P_7	IO_L27N_8	AN14	FPGA_CON330	23	H33	IO_L6P_9	IO_L27N_10	M8	FPGA_CON74	23
23	FPGA_CON395	AN32	IO_L6N_7	IO_L27P_8	AV9	FPGA_CON331	23	H34	IO_L6N_9	IO_L27P_10	U13	FPGA_CON75	23
23	FPGA_CON396	AN30	IO_L7P_7	IO_L26N_8	AW9	FPGA_CON332	23	F36	IO_L7P_9	IO_L26N_10	T13	FPGA_CON76	23
23	FPGA_CON397	AP31	IO_L7N_7	IO_L26P_8	AV12	FPGA_CON333	23	F36	IO_L7N_9	IO_L26P_10	R11	FPGA_CON77	23
23	FPGA_CON398	AK29	IO_L8P_CC_LC_7	IO_L25N_CC_LC_8	AW12	FPGA_CON334	23	C39	IO_L8P_CC_LC_9	IO_L25N_CC_LC_10	P11	FPGA_CON78	23
23	FPGA_CON399	AI29	IO_L8N_CC_LC_7	IO_L25P_CC_LC_8	AW7	FPGA_CON335	23	D39	IO_L8N_CC_LC_9	IO_L25P_CC_LC_10	G1	FPGA_CON79	23
23	FPGA_CON400	AT24	IO_L9P_CC_LC_7	IO_L24N_CC_LC_8	AV8	FPGA_CON336	23	J32	IO_L9P_CC_LC_9	IO_L24N_CC_LC_10	F1	FPGA_CON80	23
23	FPGA_CON401	AT25	IO_L9N_CC_LC_7	IO_L24P_CC_LC_8	AU11	FPGA_CON337	23	K32	IO_L9N_CC_LC_9	IO_L24P_CC_LC_10	K6	FPGA_CON81	23
23	FPGA_CON402	AW34	IO_L10P_7	IO_L23N_VRP_8	AU12	FPGA_CON338	23	G36	IO_L10P_9	IO_L23N_VRP_10	J6	FPGA_CON82	23
23	FPGA_CON403	AV34	IO_L10N_7	IO_L23P_VRN_8	AT9	FPGA_CON339	23	G37	IO_L10N_9	IO_L23P_VRN_10	G2	FPGA_CON83	23
23	FPGA_CON404	AW30	IO_L11P_7	IO_L22N_8	AT10	FPGA_CON340	23	F37	IO_L11P_9	IO_L22N_10	G3	FPGA_CON84	23
23	FPGA_CON405	AW31	IO_L11N_7	IO_L22P_8	AU13	FPGA_CON341	23	F38	IO_L11N_9	IO_L22P_10	J5	FPGA_CON85	23
23	FPGA_CON406	AV33	IO_L12P_7	IO_L21N_8	AV13	FPGA_CON342	23	H35	IO_L12P_9	IO_L21N_10	H5	FPGA_CON86	23
23	FPGA_CON407	AU33	IO_L12N_VREF_7	IO_L21P_8	AP9	FPGA_CON343	23	J35	IO_L12N_VREF_9	IO_L21P_10	L8	FPGA_CON87	23
23	FPGA_CON408	AP25	IO_L13P_7	IO_L20N_VREF_8	AR9	FPGA_CON344	23	M30	IO_L13P_9	IO_L20N_VREF_10	K7	FPGA_CON88	23
23	FPGA_CON409	AP26	IO_L13N_7	IO_L20P_8	AP12	FPGA_CON345	23	M31	IO_L13N_9	IO_L20P_10	R12	FPGA_CON89	23
23	FPGA_CON410	AT31	IO_L14P_7	IO_L19N_8	AR12	FPGA_CON346	23	F39	IO_L14P_9	IO_L19N_10	P12	FPGA_CON90	23
23	FPGA_CON411	AT30	IO_L14N_7	IO_L19P_8	AT8	FPGA_CON347	23	F39	IO_L14N_9	IO_L19P_10	F1	FPGA_CON91	23
23	FPGA_CON412	AU25	IO_L15P_7	IO_L18N_8	AU8	FPGA_CON348	23	J34	IO_L15P_9	IO_L18N_10	F2	FPGA_CON92	23
23	FPGA_CON413	AV25	IO_L15N_7	IO_L18P_8	AL13	FPGA_CON349	23	K34	IO_L15N_9	IO_L18P_10	T14	FPGA_CON93	23
23	FPGA_CON414	AR31	IO_L16P_7	IO_L17N_8	AL14	FPGA_CON350	23	F38	IO_L16P_9	IO_L17N_10	R14	FPGA_CON94	23
23	FPGA_CON415	AR32	IO_L16N_7	IO_L17P_8	AU6	FPGA_CON351	23	G38	IO_L16N_9	IO_L17P_10	D1	FPGA_CON95	23
23	FPGA_CON416	AL26	IO_L17P_7	IO_L16N_8	AU7	FPGA_CON352	23	K33	IO_L17P_9	IO_L16N_10	D2	FPGA_CON96	23
23	FPGA_CON417	AM27	IO_L17N_7	IO_L16P_8	AP11	FPGA_CON353	23	L33	IO_L17N_9	IO_L16P_10	F3	FPGA_CON97	23
23	FPGA_CON418	AU31	IO_L18P_7	IO_L15N_8	AN12	FPGA_CON354	23	H37	IO_L18P_9	IO_L15N_10	F4	FPGA_CON98	23
23	FPGA_CON419	AU32	IO_L18N_7	IO_L15P_8	AJ12	FPGA_CON355	23	H38	IO_L18N_9	IO_L15P_10	N10	FPGA_CON99	23
23	FPGA_CON420	AV28	IO_L19P_7	IO_L14N_8	AH13	FPGA_CON356	23	N30	IO_L19P_9	IO_L14N_10	M10	FPGA_CON100	23
23	FPGA_CON421	AU28	IO_L19N_7	IO_L14P_8	AN10	FPGA_CON357	23	P29	IO_L19N_9	IO_L14P_10	T15	FPGA_CON101	23
23	FPGA_CON422	AW32	IO_L20P_7	IO_L13N_8	AM11	FPGA_CON358	23	L34	IO_L20P_9	IO_L13N_10	R16	FPGA_CON102	23
23	FPGA_CON423	AV32	IO_L20N_VREF_7	IO_L13P_8	AL10	FPGA_CON359	23	L35	IO_L20N_VREF_9	IO_L13P_10	F3	FPGA_CON103	23
23	FPGA_CON424	AW25	IO_L21P_7	IO_L12P_8	AM10	FPGA_CON360	23	J36	IO_L21P_9	IO_L12P_10	F4	FPGA_CON104	23
23	FPGA_CON425	AW26	IO_L21N_7	IO_L11N_8	AV6	FPGA_CON361	23	J37	IO_L21N_9	IO_L11N_10	N12	FPGA_CON105	23
23	FPGA_CON426	AP29	IO_L22P_7	IO_L11P_8	AW7	FPGA_CON362	23	K36	IO_L22P_9	IO_L11P_10	M11	FPGA_CON106	23
23	FPGA_CON427	AN29	IO_L22N_7	IO_L10N_8	AW4	FPGA_CON363	23	L36	IO_L22N_9	IO_L10N_10	G5	FPGA_CON107	23
23	FPGA_CON428	AN27	IO_L23P_VRN_7	IO_L10P_8	AW5	FPGA_CON364	23	H39	IO_L23P_VRN_9	IO_L10P_10	F5	FPGA_CON108	23
23	FPGA_CON429	AN28	IO_L23N_VRP_7	IO_L9N_CC_LC_8	AK11	FPGA_CON365	23	J39	IO_L23N_VRP_9	IO_L9N_CC_LC_10	G6	FPGA_CON109	23
23	FPGA_CON430	AL28	IO_L24P_CC_LC_7	IO_L9P_CC_LC_8	AL11	FPGA_CON366	23	M33	IO_L24P_CC_LC_9	IO_L9P_CC_LC_10	G7	FPGA_CON110	23
23	FPGA_CON431	AM28	IO_L24N_CC_LC_7	IO_L8N_CC_LC_8	AV3	FPGA_CON367	23	N32	IO_L24N_CC_LC_9	IO_L8N_CC_LC_10	J7	FPGA_CON111	23
23	FPGA_CON432	AP27	IO_L25P_CC_SM7_LC_7	IO_L8P_CC_LC_8	AV4	FPGA_CON368	23	R28	IO_L25P_CC_LC_9	IO_L8P_CC_LC_10	H7	FPGA_CON112	23
23	FPGA_CON433	AR27	IO_L25N_CC_SM7_LC_7	IO_L7N_8	AR7	FPGA_CON369	23	R29	IO_L25N_CC_LC_9	IO_L7N_10	F6	FPGA_CON113	23
23	FPGA_CON434	AV30	IO_L26P_SM6_7	IO_L7P_8	AR8	FPGA_CON370	23	M35	IO_L26P_9	IO_L7P_10	F6	FPGA_CON114	23
23	FPGA_CON435	AU30	IO_L26N_SM6_7	IO_L6N_8	AN7	FPGA_CON371	23	N35	IO_L26N_9	IO_L6N_10	C2	FPGA_CON115	23
23	FPGA_CON436	AR26	IO_L27P_SM5_7	IO_L6P_8	AP7	FPGA_CON372	23	K37	IO_L27P_9	IO_L6P_10	C3	FPGA_CON116	23
23	FPGA_CON437	AT26	IO_L27N_SM5_7	IO_L5N_8	AN15	FPGA_CON373	23	K38	IO_L27N_9	IO_L5N_10	D5	FPGA_CON117	23
23	FPGA_CON438	AW29	IO_L28P_7	IO_L5P_8	AM15	FPGA_CON374	23	N33	IO_L28P_9	IO_L5P_10	D6	FPGA_CON118	23
23	FPGA_CON439	AV29	IO_L28N_VREF_7	IO_L4N_VREF_8	AK9	FPGA_CON375	23	N34	IO_L28N_VREF_9	IO_L4N_VREF_10	D4	FPGA_CON119	23
23	FPGA_CON440	AV27	IO_L29P_SM4_7	IO_L4P_8	AL9	FPGA_CON376	23	P31	IO_L29P_9	IO_L4P_10	C4	FPGA_CON120	23
23	FPGA_CON441	AW27	IO_L29N_SM4_7	IO_L3N_8	AN8	FPGA_CON377	23	P32	IO_L29N_9	IO_L3N_10	C5	FPGA_CON121	23
23	FPGA_CON442	AT29	IO_L30P_SM3_7	IO_L3P_8	AN9	FPGA_CON378	23	K39	IO_L30P_9	IO_L3P_10	B5	FPGA_CON122	23
23	FPGA_CON443	AR29	IO_L30N_SM3_7	IO_L2N_8	AJ9	FPGA_CON379	23	L39	IO_L30N_9	IO_L2N_10	B3	FPGA_CON123	23
23	FPGA_CON444	AU26	IO_L31P_SM2_7	IO_L2P_8	AJ10	FPGA_CON380	23	L38	IO_L31P_9	IO_L2P_10	A3	FPGA_CON124	23
23	FPGA_CON445	AU27	IO_L31N_SM2_7	IO_L1N_8	AL5	FPGA_CON381	23	M38	IO_L31N_9	IO_L1N_10	A4	FPGA_CON125	23
23	FPGA_CON446	AT28	IO_L32P_SM1_7	IO_L1P_8	AV5	FPGA_CON382	23	M36	IO_L32P_9	IO_L1P_10	A5	FPGA_CON126	23
23	FPGA_CON447	AR28	IO_L32N_SM1_7			FPGA_CON383	23	M37	IO_L32N_9			FPGA_CON127	23



8 SMA_NOE

SSRAM INTERFACE

NAND FLASH

8 NDAREADY
8 NDARE
8 NDACE
8 NDACLE
8 NDAALE
8 NDAAnWE
8 NDA_D7
8 NDA_D6
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14 EXP_IO_128

VDD3V3 AD28
VDD3V3 AE25
VDD3V3 AE32
VDD3V3 AG29
VDD3V3 AI33
VDD3V3 AM34
VDD3V3 AP38
VDD3V3 AR35
VDD3V3 AU39
VDD3V3 AV36

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AJ3 VDD3V3
AH6 VDD3V3
AG9 VDD3V3
AF12 VDD3V3

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FPGA INTERCONNECTION

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AA38

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VDD3V3 P38
VDD3V3 R35
VDD3V3 T32
VDD3V3 U29
VDD3V3 U39
VDD3V3 U39
VDD3V3 V26
VDD3V3 V36
VDD3V3 W33

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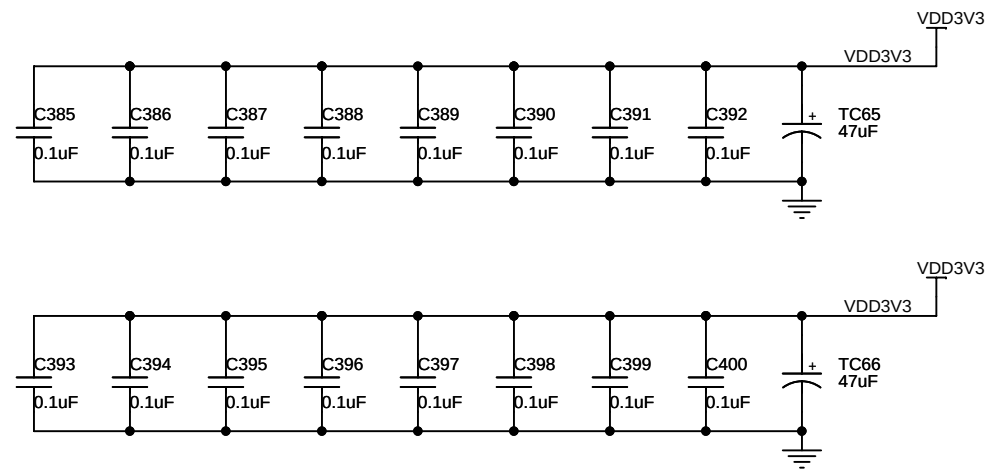
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IO_L4P_13
IO_L4N_VREF_13
IO_L5P_13
IO_L5N_13
IO_L6P_13
IO_L6N_13
IO_L7P_13
IO_L7N_13
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IO_L8N_CC_LC_13
IO_L9P_CC_LC_13
IO_L9N_CC_LC_13
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IO_L8P_CC_LC_14
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IO_L2P_14
IO_L1N_14
IO_L1P_14

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Y10 VDD3V3
W13 VDD3V3
V6 VDD3V3
U9 VDD3V3
T2 VDD3V3
R5 VDD3V3
N1 VDD3V3
M4 VDD3V3
J3 VDD3V3



VIRTEX-4 FPGA
POWER GROUND

GND D13	GND	GND J28	GND
GND K13	GND	GND N28	GND
GND M13	GND	GND W28	GND
GND P13	GND	GND AJ28	GND
GND AD13	GND	GND AW28	GND
GND AE13	GND	GND B29	GND
GND AK13	GND	GND M29	GND
GND AP13	GND	GND AB29	GND
GND G14	GND	GND AM29	GND
GND I14	GND	GND E30	GND
GND N14	GND	GND R30	GND
GND U14	GND	GND AE30	GND
GND W14	GND	GND AR30	GND
GND AE14	GND	GND H31	GND
GND AG14	GND	GND V31	GND
GND AJ14	GND	GND AH31	GND
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GND P15	GND	GND L32	GND
GND V15	GND	GND AA32	GND
GND Y15	GND	GND AL32	GND
GND AD15	GND	GND D33	GND
GND AE15	GND	GND P33	GND
GND AK15	GND	GND AD33	GND
GND C16	GND	GND AP33	GND
GND N16	GND	GND G34	GND
GND U16	GND	GND U34	GND
GND W16	GND	GND AG34	GND
GND AC16	GND	GND AU34	GND
GND AF16	GND	GND K35	GND
GND AN16	GND	GND Y35	GND
GND F17	GND	GND AK35	GND
GND T17	GND	GND C36	GND
GND V17	GND	GND N36	GND
GND AF17	GND	GND AC36	GND
GND AT17	GND	GND AN36	GND
GND J18	GND	GND E37	GND
GND W18	GND	GND T37	GND
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GND AJ18	GND	GND AT37	GND
GND AW18	GND	GND A38	GND
GND B19	GND	GND J38	GND
GND M19	GND	GND W38	GND
GND V19	GND	GND AJ38	GND
GND AB19	GND	GND AW38	GND
GND AD19	GND	GND B39	GND
GND AM19	GND	GND M39	GND
GND F20	GND	GND AB39	GND
GND R20	GND	GND AM39	GND
		GND AV39	GND
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GND AR20	GND	GND H1	GND
GND H21	GND	GND V1	GND
GND T21	GND	GND AH1	GND
GND V21	GND	GND AV1	GND
GND AB21	GND	GND A2	GND
GND AH21	GND	GND L2	GND
GND AV21	GND	GND AA2	GND
GND A22	GND	GND AL2	GND
GND I22	GND	GND AW2	GND
GND U22	GND	GND D3	GND
GND AA22	GND	GND P3	GND
GND AL22	GND	GND AD3	GND
GND D23	GND	GND AP3	GND
GND P23	GND	GND G4	GND
GND AB23	GND	GND U4	GND
GND AD23	GND	GND AG4	GND
GND AP23	GND	GND AU4	GND
GND G24	GND	GND K5	GND
GND R24	GND	GND Y5	GND
GND U24	GND	GND AK5	GND
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GND Y25	GND	GND AT7	GND
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GND AE25	GND	GND WR	GND
GND AH25	GND	GND AJ8	GND
GND AK25	GND	GND AW8	GND
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GND R26	GND	GND AM9	GND
GND AA26	GND	GND E10	GND
GND AC26	GND	GND R10	GND
GND AG26	GND	GND AE10	GND
GND AJ26	GND	GND AR10	GND
GND AN26	GND	GND H11	GND
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SSRAM INTERFACE

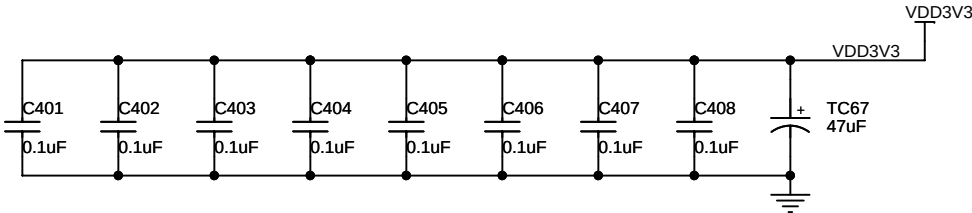
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8 SMA_A8	Y29
8 SMA_A9	AA30
8 SMA_A10	AC38
8 SMA_A11	AC39
8 SMA_A12	AA31
8 SMA_A13	Y31
8 SMA_A14	AC35
8 SMA_A15	AB35
8 SMA_A16	AB33
8 SMA_A17	AA33
8 SMA_A18	AC33
8 SMA_A19	AC34
8 SMA_A20	AD37
8 SMA_nADSC	AC37
8 SMA_nADSP	AC32
8 SMA_nADV	AB31
8 SMA_nBW1	AE39
8 SMA_nBW2	AD39
8 SMA_nBW3	AE38
8 SMA_nBW4	AE39
8 SMA_nBWE	AD35
8 SMA_nCE	AD36
8 SMA_CLK	AC30
8 SMA_nCS1	AB30
8 SMA_nGW	AE37
8 SMA_nLBO	AE38
8 SMA_D31	AB27
8 SMA_D30	AB28
8 SMA_D29	AE34
8 SMA_D28	AD34
8 SMA_D27	AD31
8 SMA_D26	AD32
8 SMA_D25	AE36
8 SMA_D24	AE36
8 SMA_D23	AJ39
8 SMA_D22	AH39
8 SMA_D21	AG37
8 SMA_D20	AG38
8 SMA_D19	AH37
8 SMA_D18	AH38
8 SMA_D17	AE34
8 SMA_D16	AE35
8 SMA_D15	AJ36
8 SMA_D14	AJ37
8 SMA_D13	AG35
8 SMA_D12	AG36
8 SMA_D11	AJ35
8 SMA_D10	AH35
8 SMA_D9	AK38
8 SMA_D8	AK39
8 SMA_D7	AM37
8 SMA_D6	AM38
8 SMA_D5	AL38
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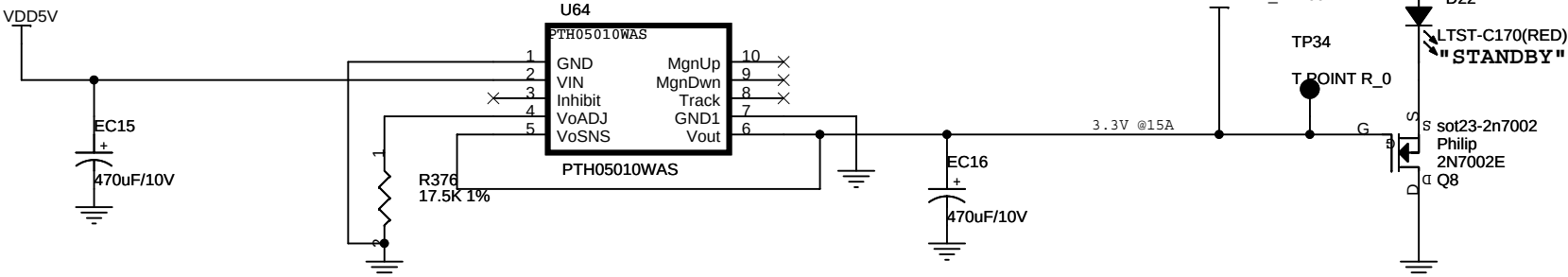
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VDD3V3 AG39	VCCO_15
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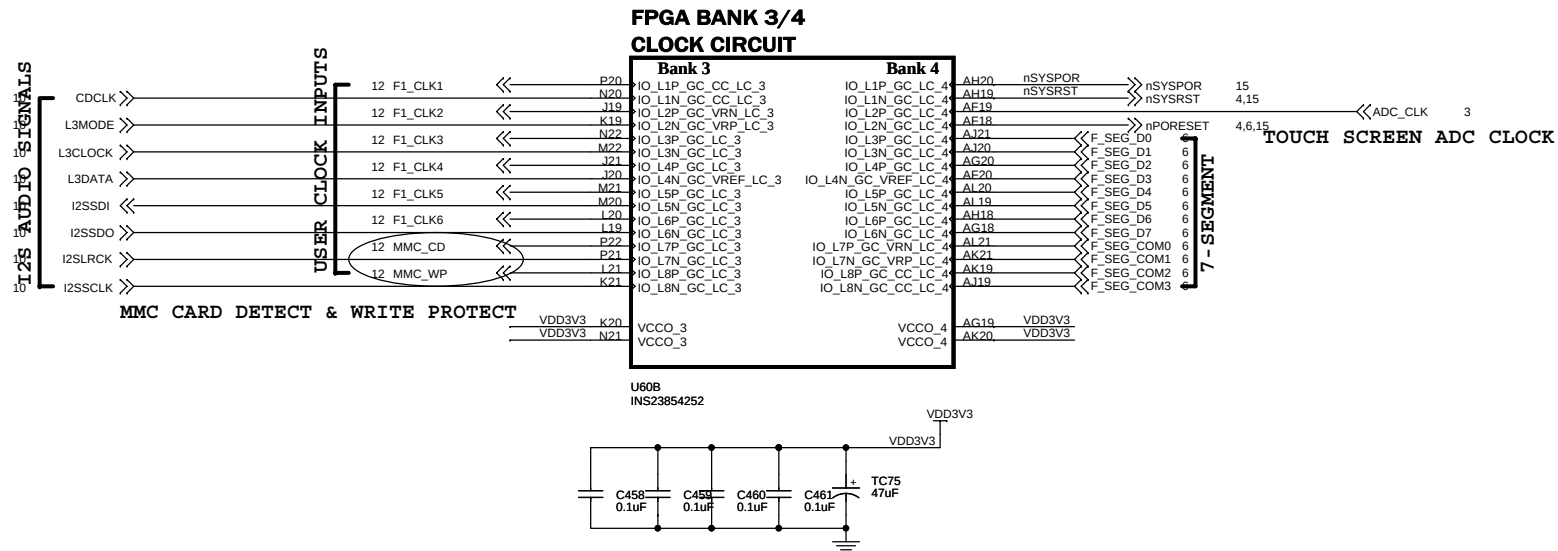
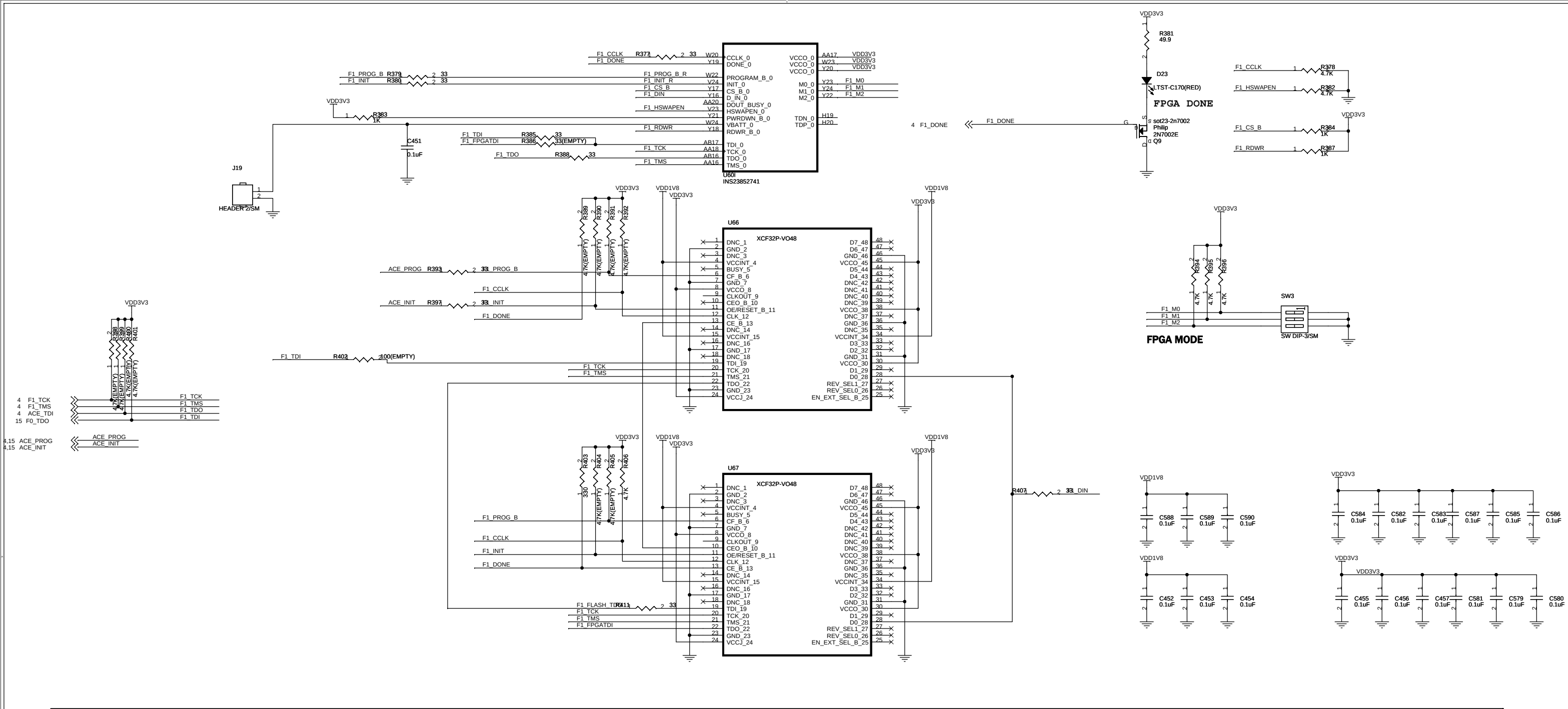
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IO_L2N_15	AF9
IO_L3P_15	AG1
IO_L3N_15	AG2
IO_L4P_15	AE5
IO_L4N_VREF_15	AE6
IO_L5P_15	AD6
IO_L5N_15	AE6
IO_L6P_15	AE4
IO_L6N_15	AE4
IO_L7P_15	AE1
IO_L7N_15	AE1
IO_L8P_CC_LC_15	AD11
IO_L8N_CC_LC_15	AC12
IO_L9P_CC_LC_15	AE2
IO_L9N_CC_LC_15	AE3
IO_L10P_15	AC7
IO_L10N_15	AD7
IO_L11P_15	AD9
IO_L11N_15	AC10
IO_L12P_15	AC14
IO_L12N_VREF_15	AB15
IO_L13P_15	AC13
IO_L13N_15	AB13
IO_L14P_15	AD4
IO_L14N_15	AD5
IO_L15P_15	AD1
IO_L15N_15	AD2
IO_L16P_15	AB8
IO_L16N_15	AC8
IO_L17P_15	AC4
IO_L17N_15	AC5
IO_L18P_15	AB10
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IO_L19P_15	AA8
IO_L19N_15	AB7
IO_L20P_15	AC2
IO_L20N_VREF_15	AC3
IO_L21P_15	AB1
IO_L21N_15	AB2
IO_L22P_15	AB5
IO_L22N_15	AB6
IO_L23P_VRN_15	AA5
IO_L23N_VRP_15	AA6
IO_L24P_CC_LC_15	AA13
IO_L24N_CC_LC_15	AA14
IO_L25P_CC_LC_15	AA1
IO_L25N_CC_LC_15	Y1
IO_L26P_15	AB3
IO_L26N_15	AA3
IO_L27P_15	Y2
IO_L27N_15	Y3
IO_L28P_15	AA4
IO_L28N_VREF_15	Y4
IO_L29P_15	Y6
IO_L29N_15	Y7
IO_L30P_15	W1
IO_L30N_15	W2
IO_L31P_15	V2
IO_L31N_15	V3
IO_L32P_15	W4
IO_L32N_15	V4

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SANGHWA FPGA PLATFORM FPGA0 POWER VCC0			
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